Jomo Kenyatta University of Agriculture and Technology

College of Engineering and Technology

School of Mechanical, Materials, and Manufacturing Engineering

Department of Mechatronic Engineering

Design and Fabrication of a Reliable Dual Recovery System for the N4 Rocket with Extended Telemetry Range

(FYP22-4)

Interim Report

Glenn Gatiba (ENM221-0149/2021)

Ian Joseph (ENM221-0249/2021)

Supervisors

Dr Shohei Aoki

March 5, 2026

Declaration

We declare that this interim report is our original work and has not been presented for examination in any other institution.

We confirm that all sources of information, data, and published material used in preparing this report have been appropriately acknowledged and referenced in accordance with university academic standards.

We further affirm that this report reflects the progress made in our Final Year Project at this stage and that the work presented is a true account of the activities completed to date.

Students:

Name: Glenn Gatiba — Date

Registration Number: ENM221-0149/2021

Name: Ian Joseph — Date

Registration Number: ENM221-0249/2021

Supervisors:

Name: Dr Shohei Aoki — Date

Department of Mechatronic Engineering

Acknowledgment

We express our sincere gratitude to our project supervisor, Dr Shohei Aoki, for his invaluable guidance, constructive feedback, and continuous support throughout this project.

We are deeply grateful to the Nakuja Project team for providing the platform, technical mentorship, and resources necessary for this work. We also appreciate our fellow team members for their collaboration and technical discussions.

We thank the Department of Mechatronic Engineering at Jomo Kenyatta University of Agriculture and Technology for providing the facilities and academic environment required to carry out this project.

Finally, we extend our appreciation to our families and friends for their encouragement and support throughout this period.

Glenn Gatiba and Ian Joseph

March 5, 2026

Abstract

The Nakuja N4 rocket recovery system addresses three failure modes identified in previous missions: insufficient telemetry range, uncontrolled nichrome-wire burnout during parachute deployment, and absence of pre-flight electrical status feedback. This interim report presents the design, analysis, and fabrication-stage results achieved to date.

The telemetry subsystem adopts the XBee-PRO 900HP operating at 900 MHz in transparent Universal Asynchronous Receiver-Transmitter (UART) mode, with a first-order link budget confirming a +32 dB margin at 5 km using omnidirectional dipole antennas. The ignition subsystem replaces the previous direct DC dump with Pulse-Width Modulation (PWM)-controlled IRLZ44N Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) switching at 1 kHz; bench characterisation of 26 AWG nichrome elements demonstrated sustained energisation for approximately 30 seconds at minimum duty cycle compared to burnout within 2 seconds at DC-equivalent drive, validating the controlled-heating approach. A battery monitoring and continuity-detection circuit centred on the ADS1115 16-bit Analog-to-Digital Converter (ADC) provides pre-launch voltage and igniter-state visibility. Mechanically, a stepped polyethylene terephthalate glycol (PETG) battery-carrier architecture accommodates four 18650 cells within the 95 mm avionics bay while routing dual threaded retention rods without interference.

At the close of this interim phase, all subsystem designs have been completed and verified through 2D/3D computer-aided design (CAD), EasyEDA printed circuit board (PCB) schematics, and Proteus circuit simulation. A custom flight-computer PCB is currently in fabrication, and 3D-printed structural parts are in production. Firmware modules for telemetry, sensor interfacing, PWM control, and flight-state management have been implemented on FreeRTOS. Pending work for the next phase includes PCB assembly, open-field XBee range testing,

hardware-level monitoring calibration, and pyrotechnic pop tests to validate deployment force.

Contents

Nomenclature

Abbreviations

ADC Analog-to-Digital Converter

AGL Above Ground Level

BMS Battery Management System

CAD Computer-Aided Design

ESP32 Espressif 32-bit Microcontroller Platform

FEA Finite Element Analysis

GPS Global Positioning System

IMU Inertial Measurement Unit

LiPo Lithium Polymer Battery

MOSFET Metal-Oxide-Semiconductor Field-Effect Transistor

PETG Polyethylene Terephthalate Glycol

PWM Pulse Width Modulation

RF Radio Frequency

UART Universal Asynchronous Receiver-Transmitter

Symbols and Units

D Duty cycle, percent

f Frequency, hertz

I Current, ampere

P Power, watt

R Resistance, ohm

T Temperature, degree Celsius

V Voltage, volt

List of Figures

List of Tables

1 Introduction

1.1 Background of Study

Rocket recovery systems represent the most critical subsystem for preserving flight data and hardware in high-power rocketry missions. The evolution of recovery systems has progressed from simple single deployment methods to sophisticated dual deployment architectures. Single deployment systems release a parachute at apogee, presenting risks of high drift radius and unpredictable landing zones. In contrast, dual deployment systems utilize a drogue parachute at apogee for stability, followed by a main parachute deployment at lower altitude for controlled soft landing, establishing the standard for modern high-power rocketry operations [14, 15].

The Nakuja Project N4 mission builds upon previous experiences from the N3.5 mission, which employed a basic single-deployment piston-based system. The current N4 design implements dual deployment architecture but faces significant challenges in deployment reliability and telemetry signal integrity. Previous missions demonstrated critical vulnerabilities in both mechanical deployment mechanisms and communication systems, necessitating a comprehensive redesign approach [1].

1.2 Problem Statement

The N4 mission's recovery system faces critical reliability challenges including telemetry range limitations causing signal degradation during critical flight phases, uncontrolled ignition system temperatures exceeding 1000 degrees Celsius that create unreliable single-use deployment mechanisms, and absence of real-time preflight monitoring systems for battery voltage and igniter continuity, collectively threatening mission success through potential vehicle loss, deployment failures, and inability to validate system readiness before launch.

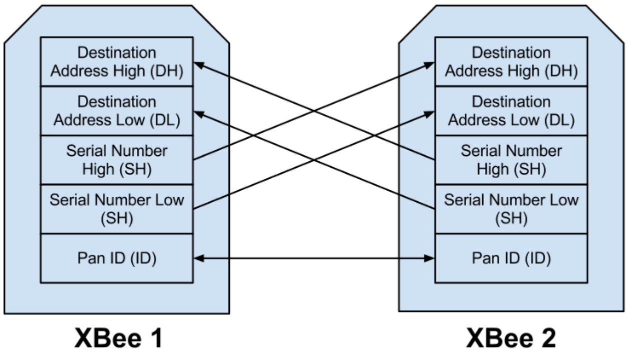


Figure 1: XBee-PRO 900HP telemetry integration architecture used to improve end-to-end link reliability

1.3 Objectives

Main Objective: To design and fabricate a reliable dual recovery system for the N4 rocket with extended telemetry range.

Specific Objectives:

1. To extend the telemetry range of the recovery system beyond 5 km.
2. To improve deployment reliability through controlled pyrotechnic ignition, targeting repeatable nichrome-wire energisation without wire burnout.
3. To design a rugged 3D-printed avionics holder with vibration isolation, battery retention, and real-time system diagnostics.
4. To validate the complete recovery system through ground and field testing.

1.4 Justification

The development of a reliable recovery system is paramount to mission success. Loss of the rocket vehicle results not only in hardware loss but also complete loss of flight data critical for mission analysis and future design improvements. The proposed improvements address documented failure modes from previous missions and implement industry-standard practices adapted for student rocketry applications.

The Nakuja Project represents a significant investment in student engineering education and practical aerospace systems development. Ensuring mission success through robust recovery systems validates this investment and provides valuable learning opportunities for future engineering professionals.

1.5 Expected Outcomes

The expected outcomes of this study are:

1. Telemetry range exceeding 5 km with reliable link performance.
2. Consistent and repeatable parachute deployment through controlled pyrotechnic ignition.
3. A fabricated 3D-printed avionics holder providing vibration isolation, battery retention, and real-time system diagnostics.
4. A validated recovery system demonstrated through ground and field testing.

2 Literature Review

2.1 Recovery Systems in Student Rocketry

Recovery systems preserve flight hardware and flight data and are therefore mission-critical in high-power student rocketry [14, 15]. A common practice is dual deployment: a drogue parachute at apogee for stability and reduced drift, then a main parachute at lower altitude for lower landing energy. Literature consistently shows that this staged strategy improves recovery reliability compared with single-deployment approaches in high-altitude missions.

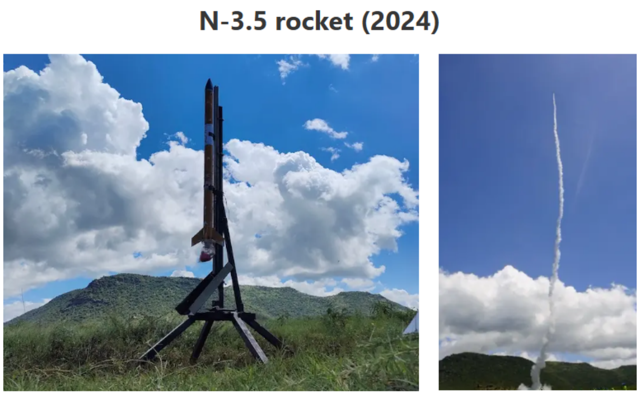


Figure 2: Full Nakuja N3 rocket showing the dual-deployment recovery configuration and component layout [1]

The Nakuja project progression from earlier single-deployment flights to N4 dual deployment reflects this broader trend and responds to observed mission-level constraints in communication and recovery robustness [1].

2.2 Telemetry Systems Analysis

Table 1 compares telemetry options reported in student rocketry and related field systems.

Comparative studies show a tradeoff between range, bandwidth, protocol relia-

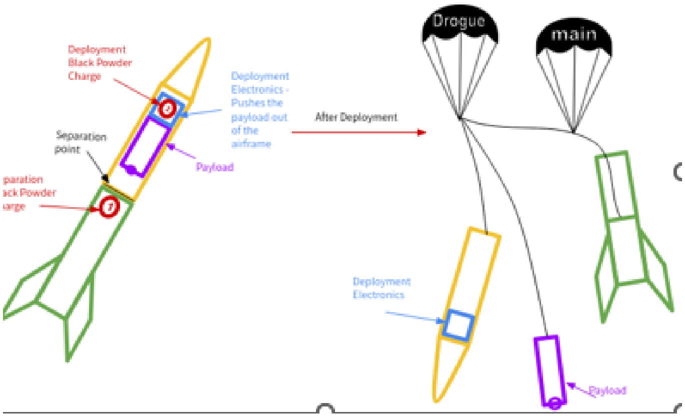


Figure 3: Rocket in flight demonstrating the dual-deployment sequence and aerodynamic stability during ascent [1]

Table 1: Telemetry System Comparison for Student Rocketry Applications

Technology	**Range**	**Data Cap**	**Strength**	**Limitation**
433 MHz RF	5–10 km	Very low	Simple, long range	No ACKs
Beacon	4 km/0.8 km	Low–Med	Simple	Range drops
LoRa SX1278	2–15 km	Low	Low power	Low BW
XBee 2.4 GHz	<2 km	Medium	Easy integration	Short range
XBee-PRO 900HP	**5–15 km**	**Med–High**	**Reliable**	**Higher power**
ESP-NOW	<1 km	Medium	Low latency	Not long-range
Cellular LTE	Global	High	Unlimited range	Network dep.

bility, and implementation complexity [16]. 900 MHz-class links are often favored for long-range recovery because of better propagation in non-ideal field conditions compared with higher-frequency short-range links [2].

Figure 4: XBee radio module used as the long-range telemetry transceiver platform [2]

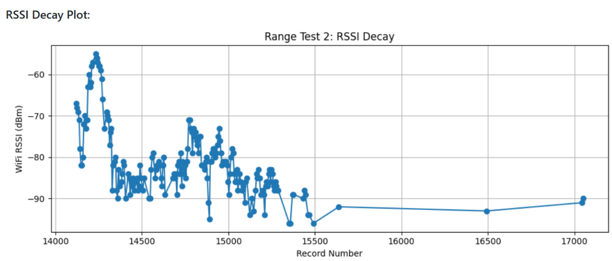


Figure 5: Beacon RSSI decay analysis showing signal degradation with distance and altitude. Authors' own data.

2.2.1 Antenna Options in Recovery Telemetry

Literature and field practice show several antenna families for rocket telemetry ground/air links: dipole and whip (omnidirectional, low complexity), patch and Bi-Quad (moderate gain, directional), and Yagi arrays (higher gain, tighter pointing requirement) [3,4]. For dynamic flight profiles, antenna pattern behavior and polarization stability are as important as nominal gain.

Figure 6: 900 MHz duck antenna used in the N4 telemetry configuration [3]

Published antenna fundamentals highlight two recurring operational issues in flight telemetry links: axis nulls in dipole-like patterns and polarization mismatch during vehicle attitude change [3]. These effects can dominate performance even when radio modules have adequate power margins.

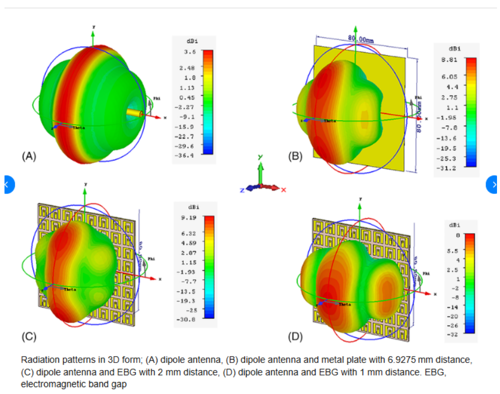


Figure 7: Dipole radiation "donut" pattern illustrating nulls along the antenna axis [4]

Antenna-related gaps from literature and field use:

- Limited student-focused procedures for selecting antenna type versus expected flight geometry.
- Limited quantified comparison of omnidirectional versus directional ground

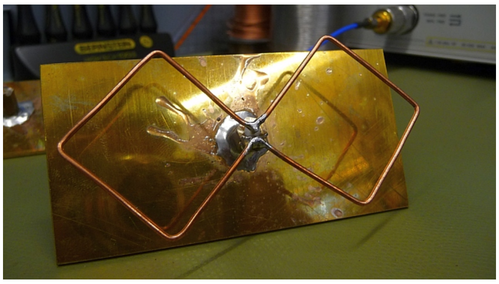


Figure 8: Example 900 MHz Bi-Quad ground-station antenna option for improved link margin [3]

antennas under real recovery scenarios.

- Inadequate treatment of polarization mismatch during descent spin/tumble in student mission workflows.

2.3 Pyrotechnic Deployment and Ignition Control

Black-powder deployment remains a common separation/ejection method in amateur and student rocketry, but reliability depends on predictable charge energy, charge containment, venting behavior, and material thermal tolerance [14, 17]. Literature also distinguishes between commercial e-match igniters and custom nichrome approaches, with tradeoffs in repeatability, cost, and field test convenience [18].

For resistive igniters, literature reports that uncontrolled energy delivery can cause premature wire failure or inconsistent ignition behavior, while controlled-energy strategies improve repeatability [18–20].

Repeated nichrome-wire pop tests in field programs commonly encounter partial melting, embrittlement, and breakage, reducing re-usability and forcing frequent replacement cycles [18, 21]. This motivates more repeatable ignition interfaces, including contained explosion-cap style arrangements that decouple ignition repeatability from wire survivability [17, 21].

Combustion literature further shows that short-duration pyrotechnic events

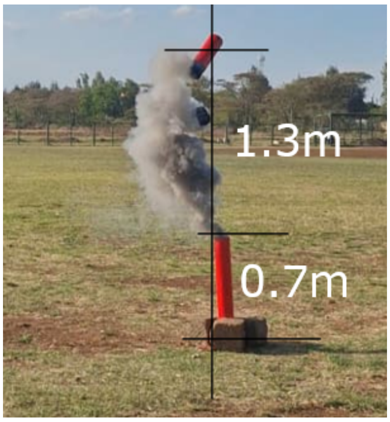


Figure 9: Excessive crimson powder deployment showing the need for precise charge control. Authors' own field test photograph [5].

Figure 10: Charge holder thermal damage from explosion highlighting thermal isolation requirements. Authors' own photograph [5].

Figure 11: Post-deployment test showing charring effects and the importance of sealed containment. Authors' own photograph [5].

Figure 12: Cut nichrome wire after pop testing, highlighting a key re-usability limitation in direct-wire ignition approaches. Authors' own photograph [5].

are often modeled as near-adiabatic transients for first-order chamber estimates, making thermal/pressure curves essential for validating containment assumptions and charge-holder boundaries [6,7]. This directly supports contained deployment architectures for safer and more predictable pop-test behavior.

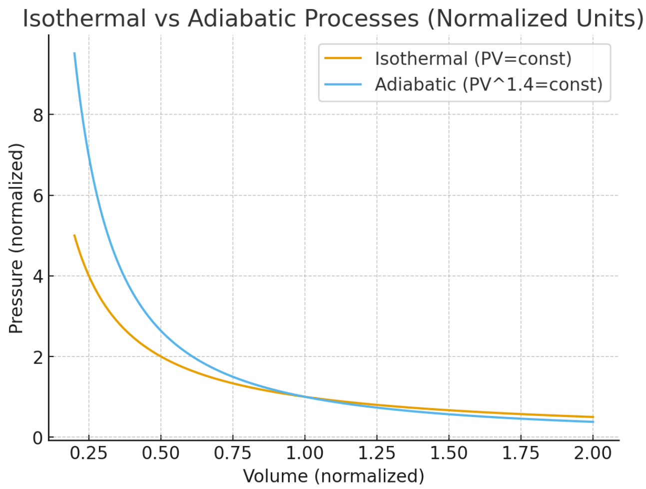


Figure 13: Normalized isothermal and adiabatic pressure-volume curves used to justify near-adiabatic explosion modelling and the need for contained deployment [6, 7]

Pyrotechnic-deployment gaps from literature and field use:

- Limited mission-specific charge mass versus ejection effectiveness data for student airframe geometries.
- Limited standardized procedures for chamber-seal verification before live deployment tests.
- Limited quantified comparison between e-match and nichrome-based systems under repeated ground-test cycles.

2.4 Power and Structural Integrity in High-G Flight

Avionics for high-power rockets must survive sustained acceleration and impulsive shock during separation/deployment. Reported failure modes include connector intermittency, battery disconnect, solder fatigue, and sensor-noise coupling [22, 23].

Launch-load theory for amateur rockets is commonly framed from thrust-generated net force and corresponding acceleration, where motor thrust characteristics directly set structural loading severity during boost [24, 25]. In addition, project pop-test records indicate short-duration pressure transients on the order of 188 kPa, reinforcing the need for mechanically robust holder designs, distributed fastening, and controlled load paths during deployment events [5].

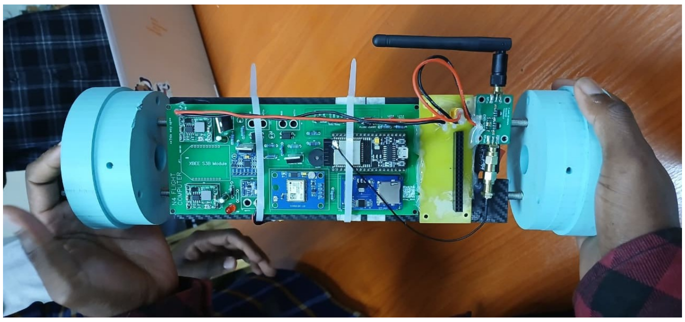


Figure 14: Assembled avionics bay with integrated battery retention and vibration isolation. Authors' own fabrication.

2.4.1 PETG for Flight-Structure Components

Recent additive-manufacturing studies show PETG as a practical compromise for functional aerospace support parts because it offers improved ductility and impact tolerance relative to more brittle alternatives, while retaining practical printability. However, structural integrity is strongly influenced by print orientation, infill/raster strategy, and inter-layer bonding quality [26–29].

This implies that material choice alone is insufficient; process-quality and test validation are essential for reliable flight-support structures.

Battery-pack integration literature and datasheets for common 18650-class cells also show non-negligible mass contribution at the module level, which increases inertial loading during launch and shock events and therefore must be included in holder-force calculations and mounting strategy design [30].

2.4.2 Flight-Computer Holder Design Progression

Literature and project practice emphasize iterative fixture refinement for avionics survivability in high-shock environments [22, 23]. Earlier holder geometries generally prioritized manufacturability but offered limited retention robustness and cable-strain management during deployment shock.

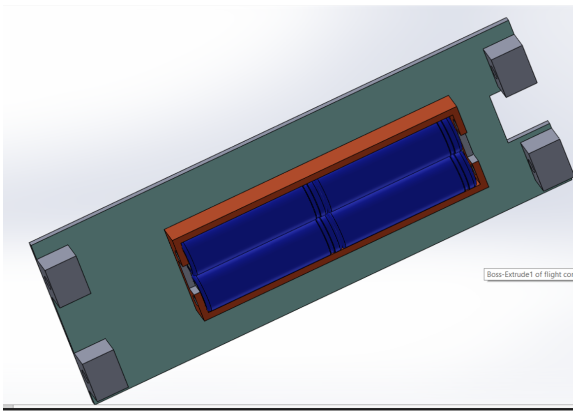


Figure 15: Old flight-computer holder design showing failure during pop tests, where paired battery-cell loading increases reaction forces on the PLA-printed holder. Authors' own design [5].

This old-holder failure observation aligns with literature recommendations to improve mechanical retention, connector stability, and shock tolerance through iterative, test-informed fixture revisions [22, 23].

2.5 Previous Nakuja Mission Evidence

Mission history indicates telemetry dropouts and deployment-side hardware stress as recurring risks in prior configurations [1]. Table 2 summarizes mission progression and key lessons.

Table 2: Nakuja Mission History and Key Findings

Mission	**Date**	**System**	**Key Outcome**
N-1	May 2021	Single deploy	Baseline established
N-2	Nov 2022	Single deploy	Telemetry improved
N-3	2024	Enhanced	Range limits identified
N-3.5	2024	Piston deploy	Thermal damage observed
N-4	**Current**	**Dual deploy**	**Development stage**

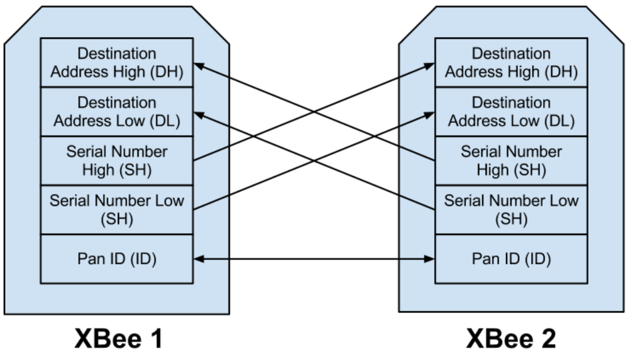


Figure 16: XBee-PRO 900HP telemetry system integration showing complete signal path [8]

2.6 Safety Standards and Reporting Practice

General rocketry safety guidance emphasizes controlled pyrotechnic handling, verified continuity before arming, and descent-rate limits compatible with payload survivability. For interim reporting practice, literature and guideline alignment also require clear separation of roles: Chapter 2 establishes state-of-the-art and gaps, while design calculations and selected implementation details are handled in methodology and results sections.

2.7 Research Gaps

The reviewed literature and Nakuja evidence identify the following unresolved gaps for N4:

1. **Telemetry antenna-geometry gap:** Limited evidence-based antenna selection and orientation guidance for student recovery missions with dynamic vehicle attitude.

2. **Pyrotechnic predictability gap:** Limited mission-specific data linking charge mass, chamber behavior, and repeatable ejection outcomes.

3. **Ignition repeatability gap:** Limited comparative data on e-match and nichrome systems under repeated pre-flight test conditions.

4. **Nichrome re-usability gap:** Limited quantified evidence on ignition consistency degradation after repeated nichrome-wire firing cycles.

5. **Containment-validation gap:** Limited use of thermochemical adiabatic-curve evidence to justify contained explosion-cap deployment designs in student workflows.

6. **Structural verification gap:** PETG support structures are common but often lack launch/deployment shock validation for mission-specific loads.

These gaps frame the need for the methodology in Chapter 3, where design calculations, selection rationale, and implementation details are developed and validated.

3 Methodology

This chapter presents the *what, why, and how* of the N4 recovery-system design process. In line with the interim-report guidelines, this chapter contains design considerations, calculations, circuit and CAD development workflow, and implementation planning for each subsystem.

3.1 Design Approach and Subsystem Decomposition

The project follows a design-build-test iteration sequence adapted from the proposal and refined for interim implementation:

1. Define subsystem-level requirements from mission constraints.
2. Generate candidate mechanical, electrical, and software concepts.
3. Select implementable concepts using performance and manufacturability criteria.
4. Build CAD and circuit designs for the selected concepts.
5. Implement embedded firmware architecture and integration interfaces.
6. Execute staged bench and system-level validation tests.

The methodology is organized into five implementation modules:

1. Telemetry link and antenna strategy,
2. Pyrotechnic deployment and ignition control,
3. Power and feedback instrumentation,
4. Mechanical containment and avionics support structure,
5. Embedded software architecture and communication control.

3.2 Telemetry Module Design Methodology

3.2.1 Telemetry Module Scope

The telemetry module covers flight-to-ground data transmission hardware, antenna strategy, communication reliability constraints, and integration interfaces to the ESP32 flight computer firmware [2, 8, 10].

3.2.2 Telemetry Design Considerations

Design considerations for telemetry selection are:

- Minimum operational range of at least 5 km line-of-sight.
- Robust packet delivery with low drop rate in dynamic flight attitude.
- Simple and deterministic integration with ESP32 UART telemetry output.
- Field-deployable ground-station setup without complex infrastructure.
- Compatibility with future communication redundancy (beacon/MQTT fallback).

3.2.3 Telemetry Parts List

3.2.4 Telemetry Design-Option Comparison

3.2.5 Chosen Telemetry Design and Justification

The selected telemetry hardware is the **Digi XBee-PRO 900HP** operating in transparent UART mode because it provides strong range performance, straightforward ESP32 integration, and robust field operation with minimal protocol complexity [2, 8, 10].

Table 3: Telemetry Module Parts List

Part	Specification	Function in Module
XBee-PRO 900HP (S3B)	900 MHz ISM, up to +24 dBm Tx power	Primary long-range telemetry radio for rocket and base station links
ESP32 Flight Computer	Dual-core MCU, UART interfaces	Generates and streams telemetry packet data to the radio interface
900 MHz Duck Dipole Antenna	Approx. +2 dBi omnidirectional gain	Baseline antenna on rocket and ground for initial long-range tests
XBee Shield/UART Interface	Transparent serial routing	Hardware adaptation between ESP32 UART and XBee module pins

3.2.6 Communication Mode Architecture and the Beacon Protocol

The N4 flight computer implements three independent communication channels managed by a software CommunicationManager [8]: (1) MQTT over WiFi for close-range pad operations, (2) the 900 MHz XBee link for primary long-range recovery telemetry, and (3) a *beacon protocol* currently being researched as a redundant fallback channel.

What is the Beacon protocol? A beacon is a raw 802.11 management frame injected directly into the IEEE 802.11 (2.4 GHz WiFi) radio medium without establishing a conventional network association. The rocket does not connect to any access point; instead, the ESP32 radio injects custom management frames — each carrying the telemetry CSV payload in a vendor-specific information element — directly into the air at a user-defined interval. A ground-station ESP32 running in promiscuous (*sniffer*) mode captures all frames that match the payload magic number regardless of association state, and extracts the RSSI directly from the hardware receive descriptor [31]. The return path (ground

Table 4: Telemetry Design Comparison Against Module Considerations

Option	**Range Capability**	**Integration Effort**	**Reliability Potential**	**Assessment vs Requirements**
XBee 2.4 GHz	Low–medium	Low	Medium	Insufficient long-range margin for target mission envelope
LoRa (generic modules)	Medium–high	Medium	Medium–high	Good range but added integration and protocol adaptation overhead
ESP-NOW/Beacon only	Medium (field dependent)	Medium	Medium	Useful backup path but not preferred as sole primary long-range channel
XBee-PRO 900HP	**High (5 km+ target)**	**Low–medium**	**High**	**Best match to range, robustness, and implementation constraints**

to rocket) uses ESP-NOW unicast packets, which provide confirmed delivery without requiring a broker or access point.

The beacon system has been field-tested to 4 km line-of-sight using a 2.4 GHz duck antenna and inline signal amplifier on the rocket side, and a directional patch antenna with amplifier at the ground station [31]. It is employed as the **redundant research channel** — the XBee 900 MHz link remains the primary mission telemetry path. The CommunicationManager supports automatic fallback from MQTT to beacon when WiFi connectivity is lost, and allows runtime switching between all three modes via serial, MQTT, or ESP-NOW commands [8].

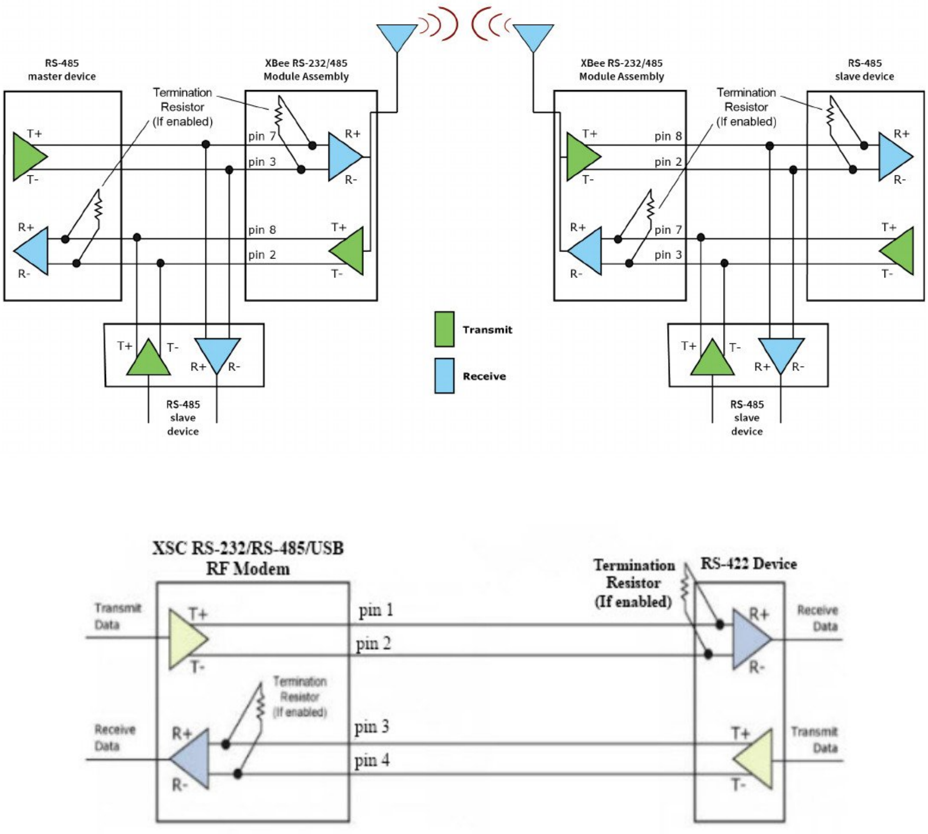


Figure 17: XBee-PRO 900HP transparent UART data-flow: ESP32 serial output routes directly through the radio link to the ground station with no packet-framing overhead [2, 9]

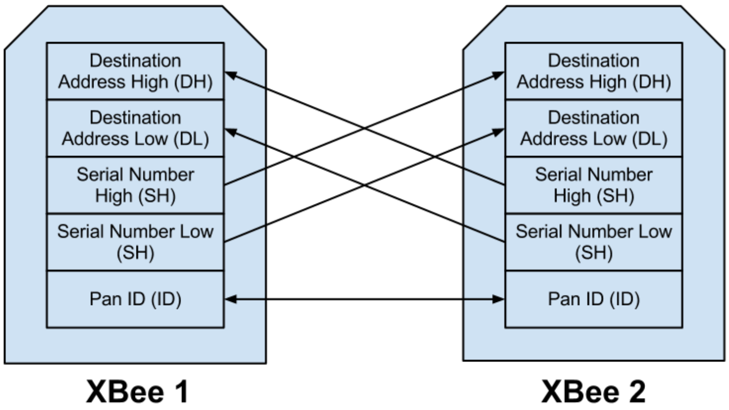


Figure 18: N4 XBee telemetry architecture: flight computer UART link to ground station [8, 10]. Authors' own diagram.

3.2.7 Antenna Options and Selection Using Link-Budget Math

Candidate antennas considered for the telemetry link are summarized in Table 5.

The selected baseline antenna pair for current implementation is the 900 MHz duck dipole. Its feasibility is checked with the first-order link budget. Free-space path loss (FSPL) models the signal attenuation over distance in the absence of obstacles [2, 4]:

$$\mathrm{FSPL}\,(\mathrm{dB}) = 20\log_{10}(d) + 20\log_{10}(f) - 147.55 \tag{1}$$

where d is the link distance in metres, f is the carrier frequency in hertz, and 147.55 is the SI-unit normalisation constant derived from the $\lambda/(4\pi R)$ free-space propagation factor [4].

Substituting the N4 target range $d = 5{,}000$ m and centre frequency $f = 915 \times 10^6$ Hz:

$$\mathrm{FSPL} \approx 105.66\,\mathrm{dB} \tag{2}$$

Received power at the ground station is obtained from the link-budget equation [2, 4]:

$$P_{rx} = P_{tx} + G_{tx} + G_{rx} - \mathrm{FSPL} \tag{3}$$

Table 5: Antenna Options for Telemetry Module

Antenna Type	**Typical Gain**	**Pattern**	**Methodology Assessment**
Duck Dipole (900 MHz)	∼2 dBi	Omnidirectional	Baseline choice due to simplicity and tolerance to vehicle rotation
Patch Antenna (900 MHz)	∼7 dBi	Directional	Higher gain but stricter pointing and build tolerance requirements
Bi-Quad (900 MHz)	∼10–11 dBi	Directional	Considered as a base-station upgrade candidate to improve null-zone robustness

where P_{tx} is the transmit power in dBm, G_{tx} and G_{rx} are the transmit and receive antenna gains in dBi, and FSPL is the free-space path loss calculated above (dB). The XBee-PRO 900HP rated output is $P_{tx} = +24$ dBm [2], and both ends use duck dipole antennas with $G_{tx} = G_{rx} = 2$ dBi:

$$P_{rx} = 24 + 2 + 2 - 105.66 \approx -77.66\,\text{dBm} \tag{4}$$

Link margin quantifies the safety buffer above the minimum detectable signal threshold [4]:

$$\text{Margin} = P_{rx} - P_{sens} \tag{5}$$

where P_{sens} is the XBee-PRO 900HP rated receiver sensitivity (−110 dBm at 10% packet-error rate [2]). Substituting the values above:

$$\text{Margin} = -77.66 - (-110) = +32.34\,\text{dB} \tag{6}$$

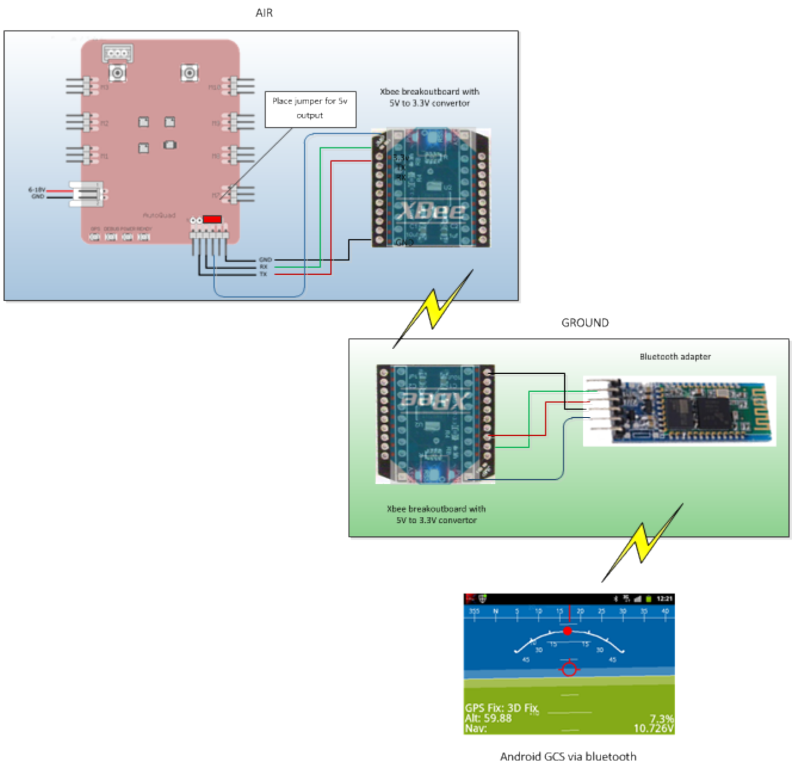


Figure 19: XBee-equipped ground control station (GCS) configuration showing transceiver, antenna, and laptop connection [11]

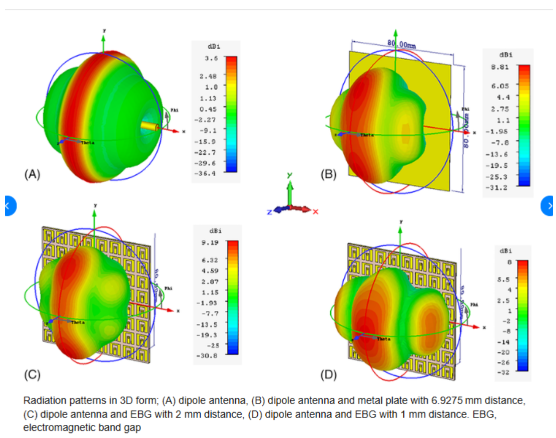


Figure 20: Dipole radiation "donut" pattern: omnidirectional main lobe with axis null zones [3, 4]

To mitigate dipole null effects at the ground segment, a **Bi-Quad antenna is being considered as a potential base-station alternative** for improved directional gain and better coverage across critical tracking sectors [3, 4].

3.3 Electrical Module with Deployment Submodule Methodology

3.3.1 Electrical + Deployment Module Scope

This module covers onboard signal interfacing, telemetry electrical routing, power/continuity monitoring, and pyrotechnic actuation electronics. It is paired with the deployment submodule because ignition-drive design depends directly on electrical interface limits and power budget constraints.

3.3.2 Design Considerations

Key electrical/deployment design considerations are:

- The flight computer has limited available pins, requiring protocol-level optimization between UART and I^2C paths.
- Telemetry radio (XBee) requires hardware serial, with remapped RX/TX pins on ESP32 to preserve other interfaces.
- Power monitoring and chute-state sensing require additional channels beyond native MCU pin allocation, motivating ADC expansion over I^2C.
- The electronics stack must fit inside the 95 mm avionics bay and align with holder mounting geometry and rod clearances.
- The deployment channel sees high-voltage transients (up to 15 V at chute lines), while BMS-limited bus operation is capped around 14 V.

- Pyro switching must be compatible with 3.3 V logic drive and sustain current pulses without excessive heating.

3.3.3 Protocol and Interface Option Comparison

Table 6: Electrical Interface Options for Pin-Constrained Flight Computer Integration

Interface Option	**Pin Usage**	**Suitability for XBee**	**Suitability for Monitoring**	**Assessment**
UART-only for all peripherals	High	Excellent for XBee stream	Poor channel scalability	Not optimal for mixed telemetry + sensing architecture
I^2C-only for all peripherals	Low	Poor for high-rate transparent radio stream	Good for ADC expansion	Not suitable as sole interface due to XBee serial requirements
Hybrid: UART + I^2C (chosen)	**Balanced**	**Best for XBee hardware serial**	**Best for ADS1115 expansion**	**Best fit for pin limits and subsystem coexistence**

3.3.4 Chosen Electrical Architecture

The selected architecture uses hardware UART for XBee telemetry and I^2C for ADS1115-based monitoring expansion. This preserves deterministic serial throughput for radio transmission while offloading analog channel scaling to an external converter integrated on the existing I^2C bus [2, 8, 10, 12, 32].

3.3.5 Dimensional Constraints and Power Requirements

Table 7: Electrical Module Constraints and Design Power Budget

Constraint/Paramete	Design Value	Implication
Avionics bay diameter envelope	95 mm	PCB and daughter-board placements constrained by cylindrical packaging
PCB footprint target	up to 70 mm × 70 mm	Allows routing margin while preserving service clearance
XBee supply rail	3.3 V (radio domain)	Requires clean low-voltage rail and UART-level compatibility
Main pyro bus	14–15 V domain	High-energy actuation path isolated from logic rail
Chute line deployment potential	up to 15 V (observed), BMS-limited to ∼14 V nominal cap	Monitoring divider and ADC scaling must tolerate both cases

3.3.6 Electrical Materials List

3.3.7 Circuit Development Workflow

Electrical implementation follows a staged concept-to-production flow:

1. Breadboard and wiring validation,
2. Fritzing-level integration drafting,
3. EasyEDA schematic and PCB capture,
4. Bench verification and revision control.

Table 8: Electrical and Deployment Submodule Materials

Component	**Key Rating/Interface**	**Role in Selected Design**
ESP32 flight computer	3.3 V logic, UART + I^2C	Core control and communication processing
XBee-PRO 900HP	UART transparent mode	Long-range telemetry radio interface
ADS1115 ADC	16-bit, I^2C interface	Voltage and chute channel expansion for pin-limited MCU
IRLZ44N MOSFET	Logic-level gate, low $R_{DS(on)}$	Pyro-channel low-side switching for PWM ignition
Voltage divider network	47 kΩ / 10 kΩ	Scales 14–15 V monitoring signals to safe ADC range
Nichrome igniter element	Resistive heating load	Converts electrical pulse to ignition heat
BMS module	Over-voltage/under-voltage protection	Caps and protects battery bus for controlled deployment operation

The Fritzing layouts below document the early wiring verification stage. Final EasyEDA schematics and fabricated PCB designs are presented in Chapter 4.

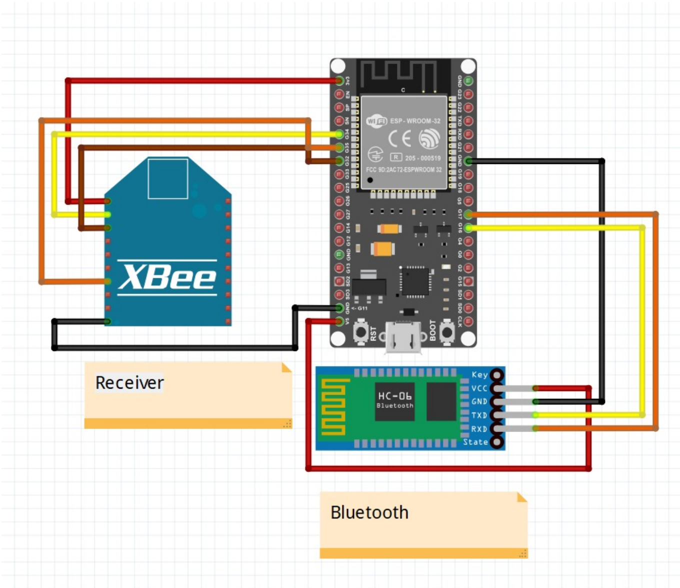


Figure 21: Base-station interconnect layout (Fritzing schematic). Authors' own diagram.

3.3.8 Power-Monitoring Calculations (ADS1115)

The monitoring path uses a 47 kΩ/10 kΩ voltage divider into the ADS1115 [12]. A voltage divider scales an input voltage proportionally using two series resistors [33]:

$$V_{ADC} = V_{in} \cdot \frac{R_2}{R_1 + R_2} = V_{in} \cdot \frac{10}{47 + 10} = V_{in} \cdot \frac{10}{57} \tag{7}$$

where V_{in} is the battery bus voltage to be measured, $R_1 = 47$ kΩ is the upper divider resistor, $R_2 = 10$ kΩ is the lower resistor, and V_{ADC} is the scaled output voltage presented to the ADC pin.

Substituting $V_{in} = 14$ V (nominal BMS-capped maximum):

$$V_{ADC,14} = 14 \cdot \frac{10}{57} \approx 2.456\,\mathrm{V} \tag{8}$$

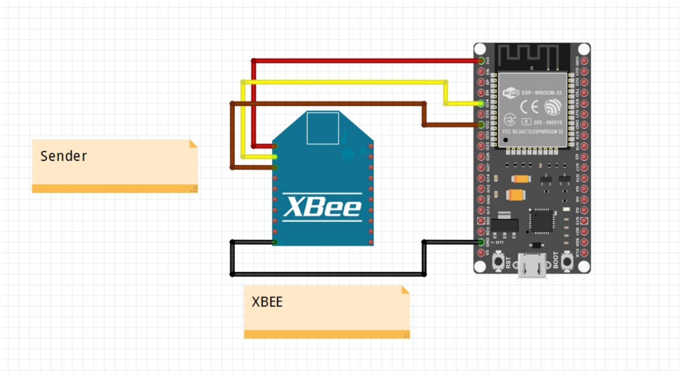


Figure 22: Rocket XBee extension wiring layout (Fritzing schematic). Authors' own diagram.

Substituting $V_{in} = 15$ V (observed chute-line deployment transient):

$$V_{ADC,15} = 15 \cdot \frac{10}{57} \approx 2.632\,\mathrm{V} \tag{9}$$

The ADS1115 is configured at a ±4.096 V full-scale range (FSR) with 16-bit resolution (32,768 counts over the positive half-range) [12]. The corresponding digital code is:

$$N = \frac{V_{ADC}}{4.096} \times 32{,}767 \tag{10}$$

where N is the 16-bit ADC output count and 32,767 is the maximum positive code ($2^{15} - 1$). Substituting the two operating points:

$$N_{14} \approx 19{,}650 \quad (\text{at } V_{in} = 14\,\mathrm{V}), \qquad N_{15} \approx 21{,}060 \quad (\text{at } V_{in} = 15\,\mathrm{V}) \tag{11}$$

These ranges provide clear monitoring margin below ADC full-scale while preserving sensitivity to deployment-state transitions.

3.3.9 Deployment PWM and MOSFET Selection

The pyro channels are driven through PWM-controlled MOSFET stages to avoid nichrome burnout and improve reusability across repeated ground tests [13, 19, 20, 34].

(a) Nichrome wire element characterisation:

The igniter elements use 26 AWG nichrome resistance wire with a linear resistance of approximately 8.5 Ω/m [18]. For a representative bridge-wire element of length $l_e = 0.10$ m (10 cm), the igniter element resistance is:

$$R_{elem} = \rho_{NiCr} \cdot l_e \tag{12}$$

where ρ_{NiCr} is the linear resistance per unit length (Ω/m) and l_e is the active element length (m). Substituting values:

$$R_{elem} = 8.5 \times 0.10 = 0.85\,\Omega \tag{13}$$

At a target drive voltage $V_{target} = 6.0$ V, steady-state current in the element is given by Ohm's Law [33]:

$$I_{elem} = \frac{V_{target}}{R_{elem}} \tag{14}$$

Substituting:

$$I_{elem} = \frac{6.0}{0.85} \approx 7.06\,\mathrm{A} \tag{15}$$

Corresponding instantaneous resistive power dissipation:

$$P_{wire} = \frac{V_{target}^2}{R_{elem}} \tag{16}$$

Substituting:

$$P_{wire} = \frac{6.0^2}{0.85} \approx 42.4\,\mathrm{W} \tag{17}$$

Using PWM to limit the effective voltage below the ignition threshold until arm state is confirmed prevents unintended heating on the bench. Bench characterisation with 26 AWG elements showed that at minimum PWM duty cycle the wire withstood approximately 30 seconds of sustained energisation without failure, compared to burnout within 2 seconds at maximum (DC-equivalent) duty.

Pyrotechnic ignition and pop tests have not yet been conducted and are scheduled for the next project phase.

(b) PWM duty-cycle model:

PWM controls the average voltage seen by the nichrome wire by rapidly switching the MOSFET on and off. The average output voltage is proportional to the duty cycle [13, 20]:

$$D = \frac{V_{target}}{V_{supply}} \tag{18}$$

where D is the dimensionless duty cycle (0 to 1), V_{target} is the desired average element voltage, and V_{supply} is the LiPo supply voltage. For the nominal case $V_{target} = 6.0$ V and $V_{supply} = 15.0$ V:

$$D = \frac{6.0}{15.0} = 0.40 \quad \text{(i.e. 40\% duty cycle)} \tag{19}$$

(c) IRLZ44N vs IRF-type MOSFET choice:

The selected switch is IRLZ44N (logic-level gate-drive compatible) rather than a standard IRF-series part, because the ESP32 gate output is 3.3 V and logic-level devices achieve low on-state resistance at reduced gate voltage [35, 36]. Conduction loss in a MOSFET switch is [36]:

$$P_{cond} = I^2 R_{DS(on)} \tag{20}$$

where I is the drain current and $R_{DS(on)}$ is the drain-to-source on-resistance. The IRLZ44N is rated $V_{DSS} = 55$ V, $I_D = 47$ A, with $R_{DS(on)} = 22$ mΩ at $V_{GS} = 5$ V [35]. For a representative $I = 3$ A pulse through a logic-level switch ($R_{DS(on)} \approx 0.022\,\Omega$):

$$P_{cond,IRLZ} = 3^2 \times 0.022 \approx 0.20\,\text{W} \tag{21}$$

For a standard non-logic-level part at low gate drive ($R_{DS(on)} \gtrsim 0.05\,\Omega$):

$$P_{cond,IRF} \gtrsim 3^2 \times 0.05 \gtrsim 0.45\,\text{W} \tag{22}$$

The lower dissipation and reliable 3.3 V gate compatibility motivate selecting IRLZ44N for the deployment channels.

3.3.10 Additional Implementation Details

To complete the electrical-deployment pairing, the design also includes:

- Hardware serial pin remapping for XBee RX/TX on ESP32 without blocking I^2C monitoring.
- Segregated low-voltage logic and high-energy pyro routing on PCB for reduced coupling.
- Pre-arm validation of battery/chute channels before enabling deployment PWM outputs.

3.4 Structural (Mechanical) Module Design Methodology

3.4.1 Structural Module Scope

The structural module defines the flight-computer holder, battery-carrier geometry, threaded-rod integration, mounting-hole allocation, and interface provision for the XBee extension board and battery management system (BMS). Beyond individual component retention, the holder functions as the primary mechanical bed that fastens the complete battery pack as a unified assembly and provides the upper-platform mounting surface for the XBee extension board and beacon antenna module.

3.4.2 Structural Design Considerations

The structural design constraints are:

- Fit within the 95 mm avionics-bay envelope.
- Survive launch and deployment shock/vibration events (design baseline: up to 4g sustained/inertial loading, with higher short-duration pop-test transients).

- Package four 18650 cells in parallel while avoiding rod interference.
- Maintain two threaded rods at 50 mm spacing and approximately 6 mm offset from the rear holder reference edge.
- Raise edge battery seats relative to mid seats so outer cells clear rod paths.
- Allocate and distribute self-tapping screw locations for load sharing.
- Provide mounting provision above battery holders for XBee extension and BMS.
- Include threaded holes for brass columns used to mount the flight computer.

3.4.3 Structural Parts List

3.4.4 Structural Design Option Comparison

3.4.5 Chosen Structural Design and Implementation

The selected design is a stepped battery-holder architecture with raised edge supports, shorter mid supports, dual threaded-rod retention, and distributed self-tapping screw points. This combination provides geometric clearance and better load sharing under launch and pop-test disturbances. The holder additionally fastens the complete 4-cell battery pack as a single retained unit, and the upper platform above the battery carriers serves as the mounting bed for both the XBee extension board and the beacon antenna module, keeping all avionics electronics within the 95 mm bay envelope.

3.4.6 Structural Load Calculations

(a) 18650 battery mass and weight Using a representative 18650 cell mass of $m_{cell} \approx 0.047$ kg (typical of high-capacity 18650 datasheets [30]), total pack mass is:

$$m_{pack} = n_{cells} \cdot m_{cell} \tag{23}$$

Table 9: Structural Module Parts List

Part	**Nominal Quantity**	**Function in Structural Module**
Flight-computer holder body (PETG)	1	Primary load-carrying printed structure supporting electronics and battery fixtures
18650 cells	4	Parallel energy source mounted in dedicated mid/edge battery carriers
Mid battery holders (shorter)	2	Support central cells while preserving assembly height and alignment envelope
Edge battery holders (raised)	2	Raise outer cells to clear threaded-rod trajectories
Threaded rods	2	Structural retention members at 50 mm spacing and 6 mm rear offset reference
Self-tapping screws	multiple	Distributed fastening points for force sharing and anti-vibration retention
Brass columns (standoffs)	4	Flight-computer mounting through threaded holes in the holder
XBee extension mount + BMS mount area	1 each	Allocated upper-plane features for electronics integration above cell carriers

Table 10: Structural Design Comparison Against Mechanical Requirements

Option	**Envelope Fit (95 mm)**	**Shock/Vibrati Robustness**	**Assembly/Inte**	**Assessment**
Single-level battery plane	Good	Medium–low	Simple	Rod interference risk for edge cells; weaker shock retention
Adhesive-only retention	Good	Low	Simple initially	Poor serviceability and limited repeatability under vibration/pop tests
Stepped holders + rods + screws	**Good**	**High**	**Moderate**	**Best balance of fit, force distribution, maintainability, and subsystem integration**

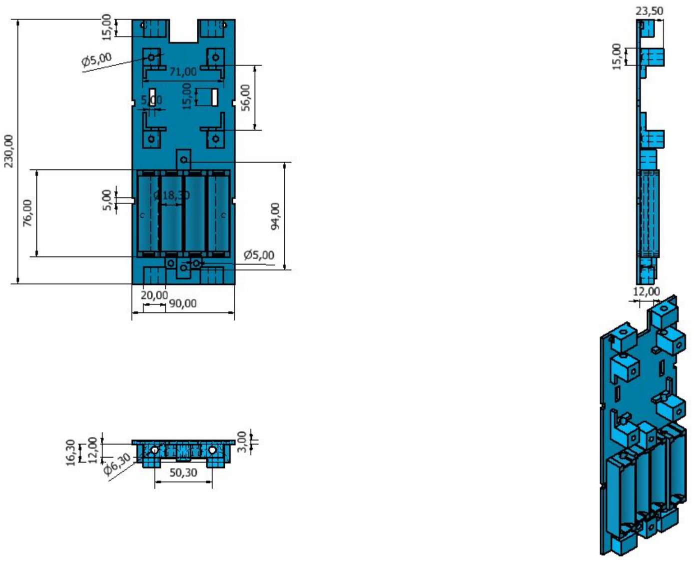


Figure 23: Flight-computer holder sled: stepped battery seating, load path, and upper electronics mounting platform

The sled fastens the complete 4-cell pack as a unified assembly and serves as the mounting bed for the XBee extension board and beacon antenna above the battery carriers.

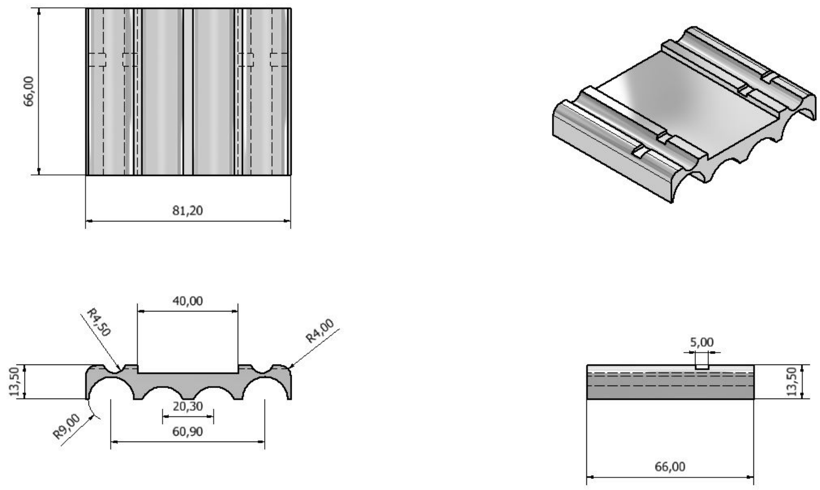


Figure 24: 18650 battery case for the stepped carrier assembly

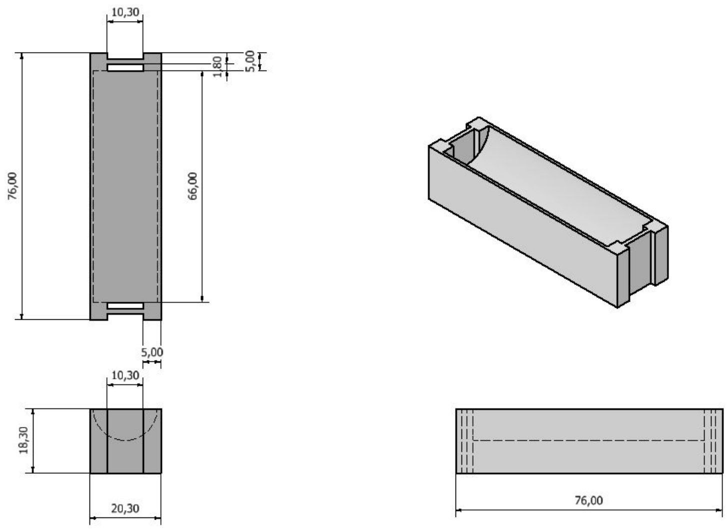


Figure 25: Raised edge battery support clearing the threaded-rod path

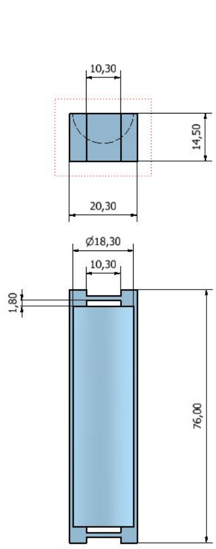


Figure 26: Mid battery support for controlled pack height and fit

where $n_{cells} = 4$ is the number of cells. Substituting:

$$m_{pack} = 4 \times 0.047 = 0.188\,\text{kg} \tag{24}$$

Static gravitational load from the four cells:

$$W_{pack} = m_{pack} \cdot g \tag{25}$$

where $g = 9.81$ m/s^2 is the standard gravitational acceleration. Substituting:

$$W_{pack} = 0.188 \times 9.81 \approx 1.84\,\text{N} \tag{26}$$

(b) Launch inertial force (4g design case) The inertial load on the battery pack during launch is obtained from Newton's second law [24, 25]:

$$F_{inertial} = m_{pack} \cdot a_{launch} \tag{27}$$

where a_{launch} is the peak launch acceleration. For a 4g design envelope:

$$a_{launch} = 4g = 4 \times 9.81 = 39.24\,\text{m/s}^2 \tag{28}$$

Substituting:

$$F_{launch} = 0.188 \times 39.24 \approx 7.38\,\text{N} \tag{29}$$

(c) Pop-test pressure-driven force (188 kPa reference) Pressure force on a surface is [5]:

$$F = P \cdot A \tag{30}$$

where P is the applied pressure (Pa) and A is the projected area (m^2). The previously recorded pop-test overpressure reference is $P_{pop} = 188$ kPa [5]. The conservative projected area uses the full 95 mm bay section:

$$A_{95} = \pi \left(\frac{0.095}{2}\right)^2 \approx 7.09 \times 10^{-3}\,\text{m}^2 \tag{31}$$

Substituting:

$$F_{pop} = P_{pop} \cdot A_{95} = 188{,}000 \times 7.09 \times 10^{-3} \approx 1.33 \times 10^3\,\text{N} \tag{32}$$

(d) Impulse estimate for pop transient Impulse is the product of force and the duration over which it acts:

$$J = F_{pop} \cdot \Delta t \tag{33}$$

where J is impulse (N·s) and Δt is the pressure-pulse duration. For a representative short pressure-pulse duration $\Delta t = 10$ ms:

$$J \approx 1{,}330 \times 0.01 \approx 13.3\,\mathrm{N} \cdot \mathrm{s} \tag{34}$$

(e) Load distribution through fasteners and supports For distributed fastening with n screw/support points, first-order load per point under a total applied force F is:

$$F_{point} = \frac{F}{n} \tag{35}$$

where n is the number of fastener points. Under $F_{pop} \approx 1{,}330$ N: 4 points carry ~ 332 N each; 8 points carry ~ 166 N each. This demonstrates why distributed fastening reduces local stress concentration and improves structural reliability.

3.5 Embedded Software Methodology

3.5.1 Software Module Scope

The software module governs all firmware executing on the ESP32 flight computer, covering three implementation areas: (i) telemetry and XBee communication management, (ii) pyrotechnic deployment state-machine control via PWM, and (iii) power and continuity monitoring integration. Area (iii) is in active development and has not yet been validated against live hardware at the time of this report.

3.5.2 Software Design Considerations

Three core problems drive the software design:

1. **XBee Integration:** The XBee-PRO 900HP must stream telemetry reliably at a deterministic data rate using hardware serial. Interface selection (SPI vs UART), data rate constraints, and pre-flight radio configuration via XCTU must all be resolved before firmware integration.

2. **Pyrotechnic Deployment via PWM:** The deployment state machine must safely arm, fire, and confirm chute deployment events using PWM-gated MOSFET drive without false actuation or premature triggering under vibration or electromagnetic interference.

3. **Power Monitoring Integration:** Real-time battery voltage and chute-channel continuity feedback from the ADS1115 must be incorporated into pre-flight arm logic and appended to the outbound telemetry packet. This integration is not yet complete in the current firmware build.

3.5.3 XBee Communication Interface: SPI vs UART

The XBee-PRO 900HP hardware exposes only a UART serial interface for transparent-mode data transfer; it does not offer SPI or I^2C bus modes [2]. SPI is therefore inapplicable, and the design uses the ESP32 hardware UART peripheral with remapped RX/TX GPIO pins, preserving all other peripheral assignments on the flight computer [32].

Key serial parameters governing the XBee-to-ESP32 link are listed in Table 11.

3.5.4 Radio Pre-Flight Configuration via XCTU

XCTU is Digi's official configuration and test utility for XBee modules. The N4 pre-flight procedure programs both flight and ground-station units via XCTU before each integration cycle [2]:

1. **Network ID (ID):** Set identically on both units (e.g., `7FFF`) to isolate the N4 network from other XBee traffic in the field.

Table 11: XBee-PRO 900HP Serial Interface Parameters

Parameter	**Value**	**Implementation Note**
Default baud rate	9600 baud	Factory default; must be updated via XCTU `BD` command
N4 firmware baud rate	115200 baud	Reduces per-packet latency; set identically on flight and ground units
Over-air RF data rate	200 kbps	Fixed by module hardware; not configurable from firmware side
Interface type	UART, 3.3 V logic	No SPI path; transparent mode passes raw bytes directly to/from ESP32
Telemetry packet interval	100 ms (10 Hz)	Frame rate used in N4 build; well below over-air throughput ceiling

2. **Baud Rate (BD):** Set to 115200 on both units to match firmware UART initialization.

3. **Channel Mask:** 900 MHz ISM band frequency-hopping profile configured to minimize ambient interference.

4. **Transmit Power (PL):** Maximum (+24 dBm) for full downlink link-margin.

5. **API Mode (AP):** Set to 0 (transparent mode) so the module passes raw bytes directly without packet-framing overhead, which suits the simple CSV/binary telemetry stream.

6. **Sleep Mode (SM):** Disabled; module remains active continuously during all flight phases.

After configuration, a loopback range test confirms bidirectional packet delivery before avionics bay assembly. Module profiles are saved in XCTU for rapid re-flashing if hardware replacement is required before launch.

3.5.5 Communication Manager Architecture

All three channels are coordinated by the CommunicationManager class, which selects and combines active transmission modes at runtime [8]. Figure 27 shows the mode-selection logic and fallback flow.

3.5.6 Firmware State Machine Architecture

The firmware is structured as a finite-state machine (FSM) executed inside the ESP32 main `loop()`, with telemetry transmission running continuously across all states via hardware serial interrupt.

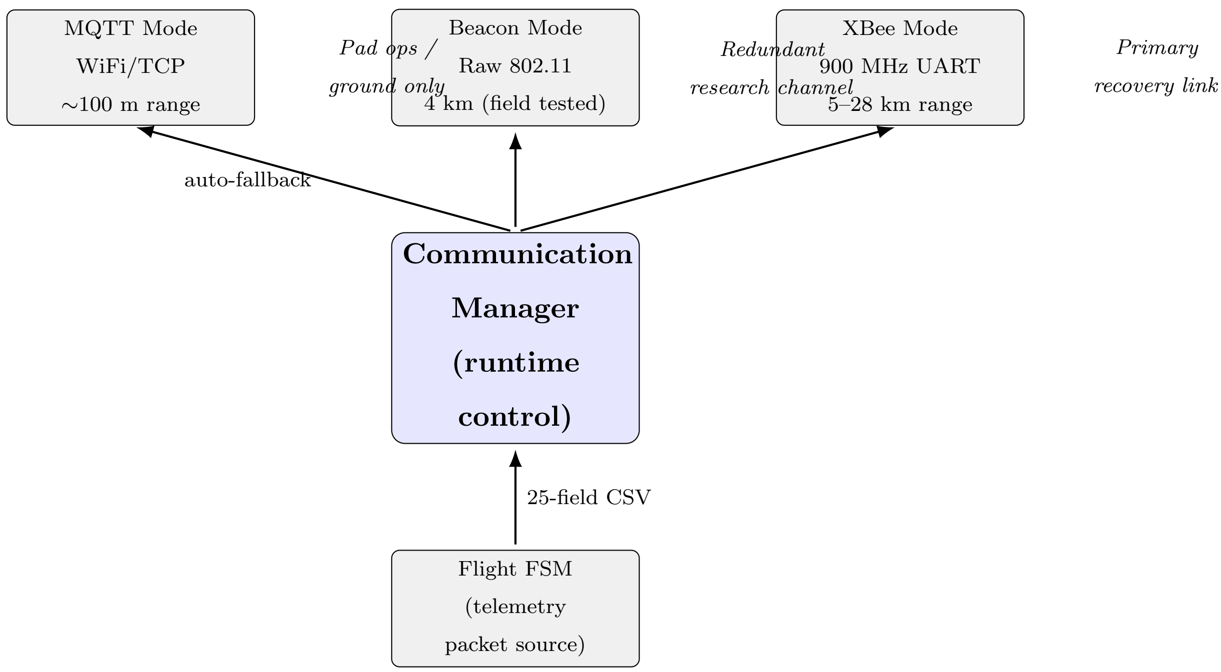


Figure 27: Communication Manager architecture: three independent physical channels selectable at runtime; XBee is the primary recovery link, Beacon is the redundant research channel [8]. Authors' own diagram.

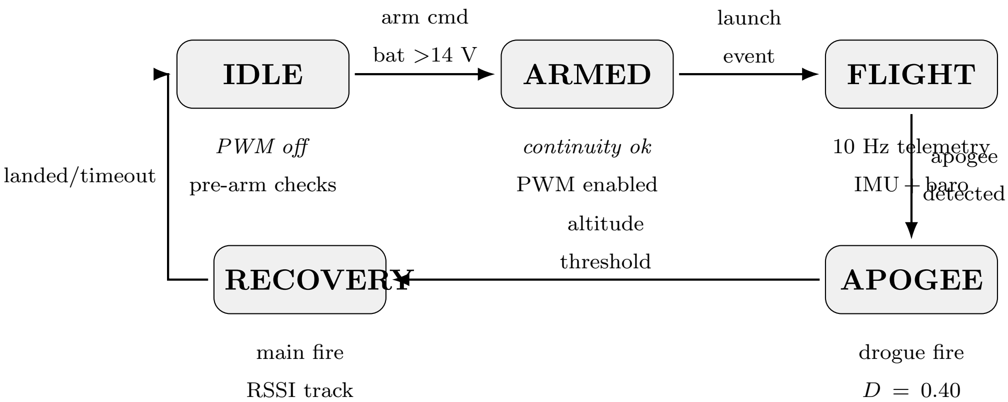


Figure 28: N4 flight firmware state machine: operational states, transitions, and per-state behaviour

3.5.7 Pyrotechnic Deployment Software Flow

The deployment channel is driven only when the FSM reaches the APOGEE or RECOVERY state and all pre-fire checks pass. Figure 29 shows the gating logic.

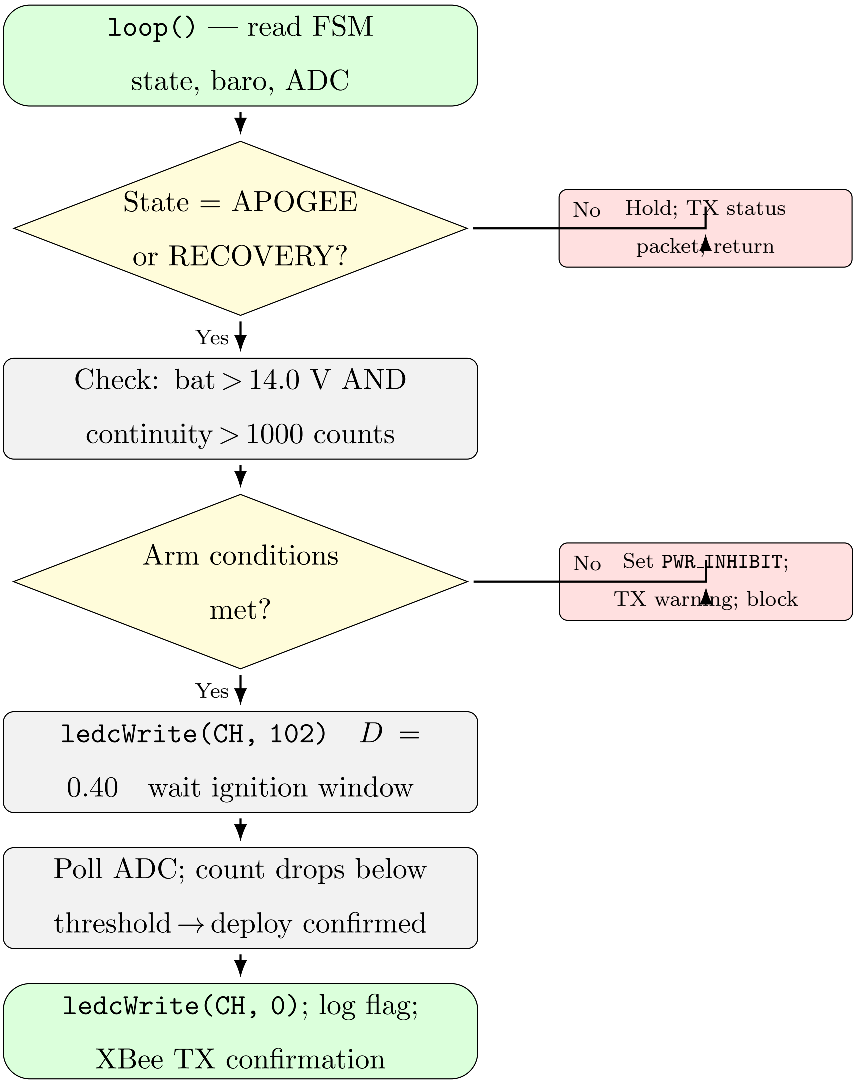


Figure 29: Deployment channel firmware flow: arm checks, PWM fire command, and post-event confirmation

Key firmware constants for the deployment channel [13, 34]:

- `PWM_DUTY_IGNITE = 102` ($D = 0.40$, $V_{\text{eff}} \approx 6$ V from 15 V supply)
- `LAUNCH_INHIBIT_V = 14.0` V minimum battery voltage for arm enable

- `CHUTE_COUNT_THRESHOLD = 1000` ADS1115 ADC counts for continuity present
- `PWM_FREQ = 1000` Hz LEDC timer frequency for the ignition channel

3.5.8 Power Monitoring Software Integration (In Progress)

Battery voltage and chute-channel continuity are read from the ADS1115 over I^2C and back-calculated to real-world voltage using the inverse of the hardware voltage divider [12]:

$$V_{battery} = V_{ADC} \times \frac{R_1 + R_2}{R_2} = V_{ADC} \times \frac{57.0}{10.0} \tag{36}$$

where V_{ADC} is the raw ADC reading in volts, and the factor 57/10 is the reciprocal of the hardware divider ratio [33].

Figure 30 shows the firmware flow for the monitoring integration.

The reconstructed voltage is appended to the telemetry packet and compared against the inhibit threshold. If the value falls below `LAUNCH_INHIBIT_V`, the deployment arm flag is cleared and a “`PWR INHIBIT`” status byte is broadcast over XBee to the ground station before launch [13].

Current integration status: The hardware design and Proteus simulation of the ADS1115 monitoring circuit are complete (detailed in Section 3, Electrical Module). Firmware I^2C polling of ADS1115 channels and the voltage reconstruction function are coded but have not yet been validated against the assembled PCB. Integration testing is planned as the next milestone before launch readiness review.

3.6 Validation and Test Method

Testing is staged from component to full-system level:

- **Component level:** PWM-output verification, continuity checks, and ADC calibration.

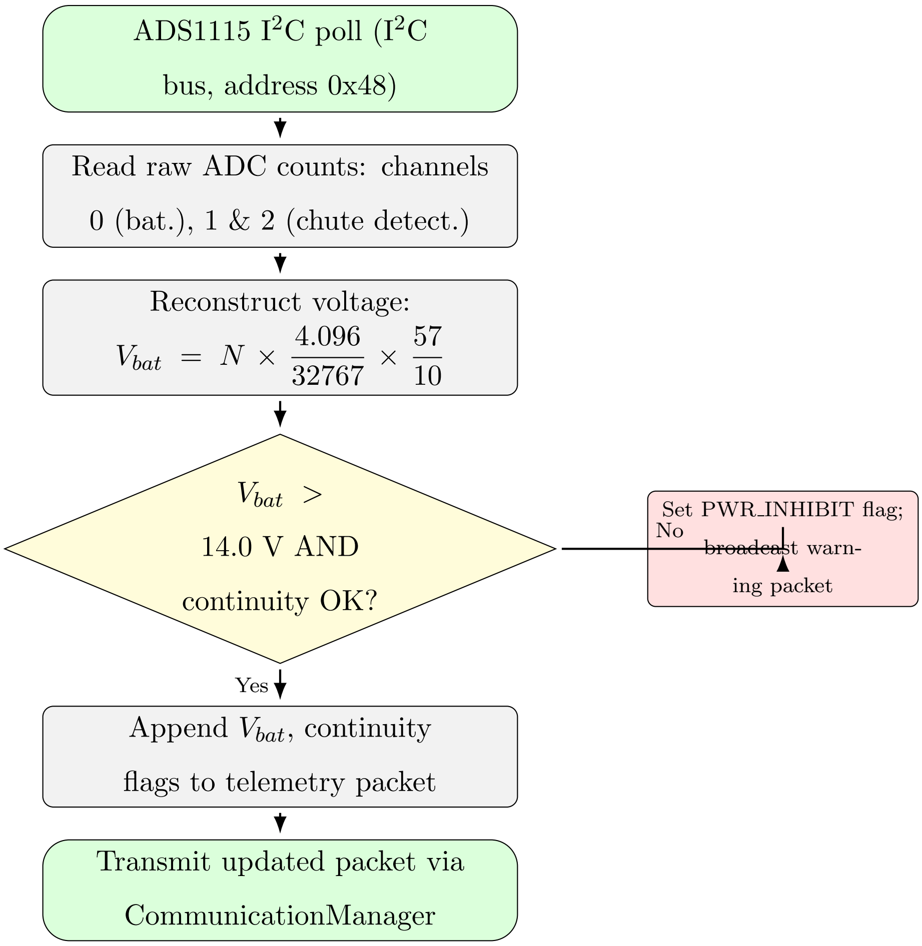


Figure 30: Power monitoring firmware flow: ADS1115 I^{2}C readout, voltage reconstruction, arm-inhibit decision, and telemetry packet update [12,13]. Authors' own diagram.

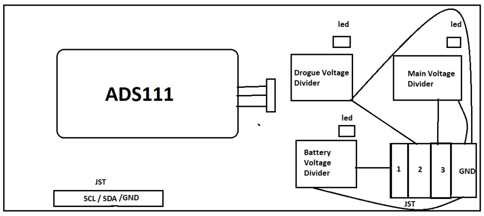


Figure 31: Power monitoring hardware concept: ADS1115 I^{2}C polling, voltage-divider network, and launch-inhibit logic [12, 13]. Authors' own diagram.

- **Subsystem level:** telemetry range trials, ignition repeatability tests, and enclosure leak/charring checks.
- **System level:** integrated ground rehearsal with full communication stack and deployment command-path validation.

4 Results and Discussion

This chapter presents the results obtained from design, fabrication, and testing activities completed to date. Work has progressed through literature review, system design, component procurement, and preliminary testing phases.

4.1 System Design Completion

4.1.1 Mechanical Structural Design

The structural design follows the guideline requirement of presenting dimensioned 2D drawings first, followed by 3D CAD models and assembly views.

2D Dimensioned Drawings Figures 32–34 present the dimensioned 2D orthographic views for the battery carrier components. All units are in millimetres.

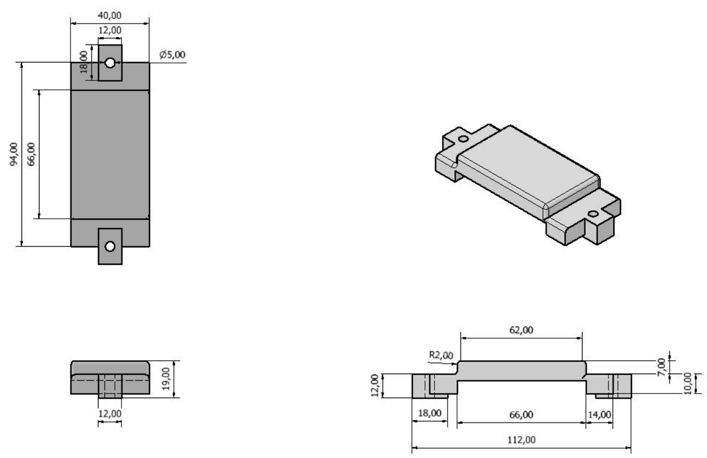


Figure 32: 18650 battery case: front, end, and top views with dimensions. Authors' own CAD.

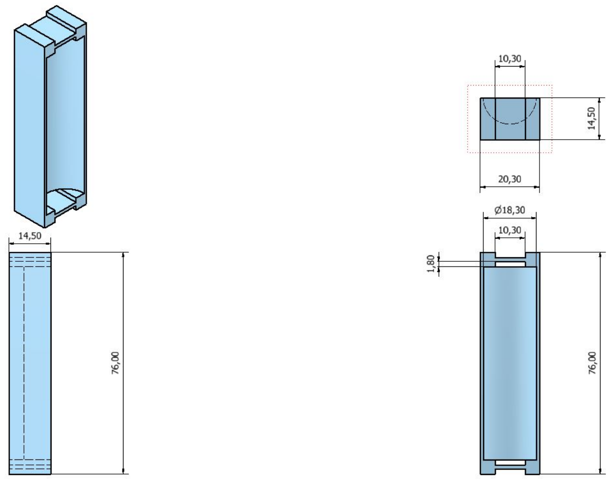


Figure 33: Mid battery support: orthographic views with key dimensions. Authors' own CAD.

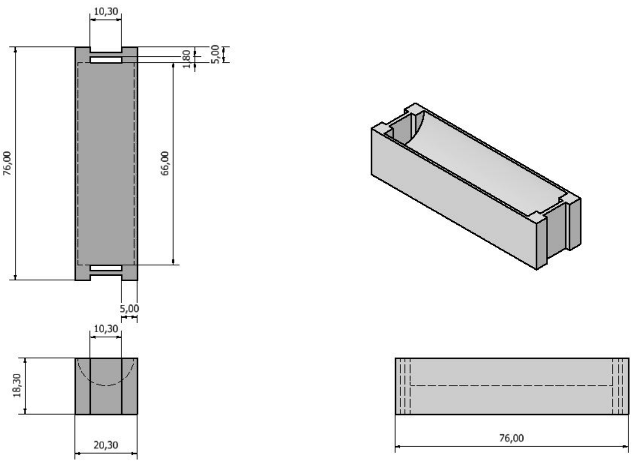


Figure 34: Raised edge battery support: orthographic views with key dimensions. Authors' own CAD.

3D CAD Models and Assembly The 3D CAD models confirm dimensional fit, threaded-rod routing, and sub-assembly relationships within the 95 mm avionics bay.

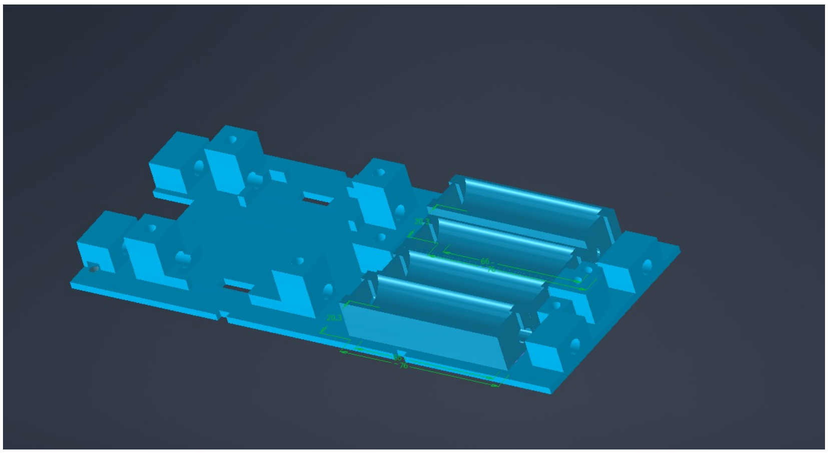


Figure 35: Flight-computer holder sled: 3D CAD model showing battery seating, rod clearances, and upper mounting platform. Authors' own CAD.

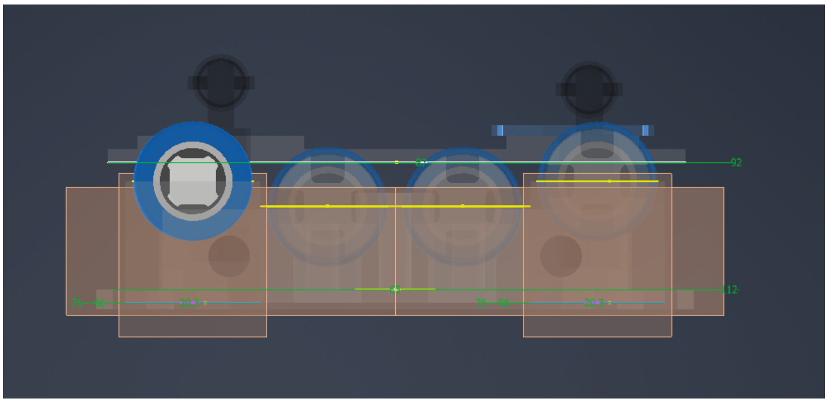


Figure 36: Threaded-rod integration: raised edge cells preserve rod clearance and structural alignment. Authors' own CAD.

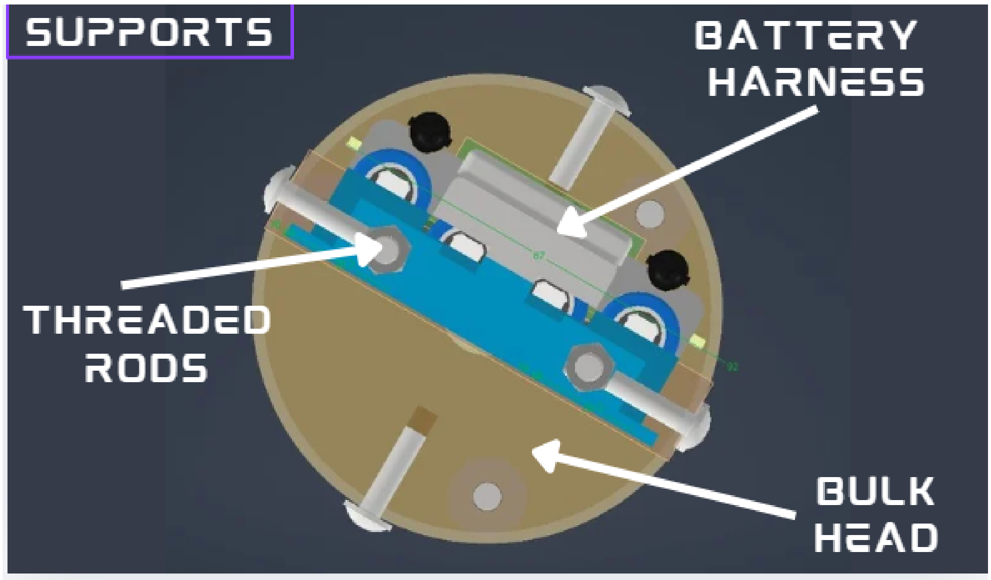
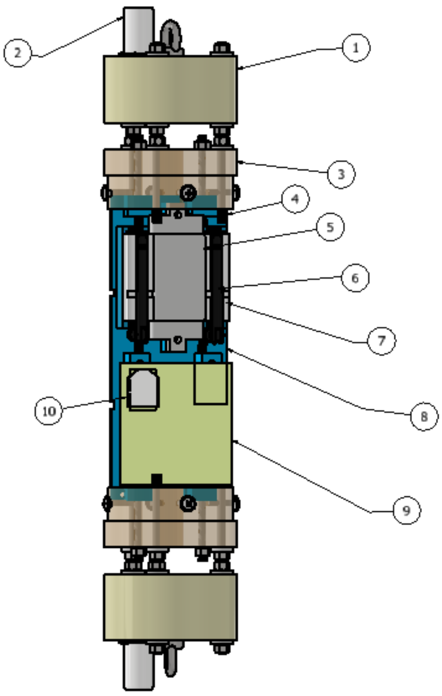


Figure 37: Labelled cross-sectional view of the avionics bay showing component stacking, retention rod paths, and internal clearances. Authors' own CAD.



PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	2	Coupler	
2	2	Charge Holders	
3	3	Bulk_Head	
4	2	Threaded Rods	
5	1	Battery Harness	STEP AP242 Edition 2
6	2	Antenna	
7	1	Battery cover	
8	1	FC Sled	
9	1	Xbee Extension Circuit	
10	1	xbee	

Figure 38: Full avionics bay assembly parts list showing all structural and electronic sub-components and their relative positions. Authors' own CAD.

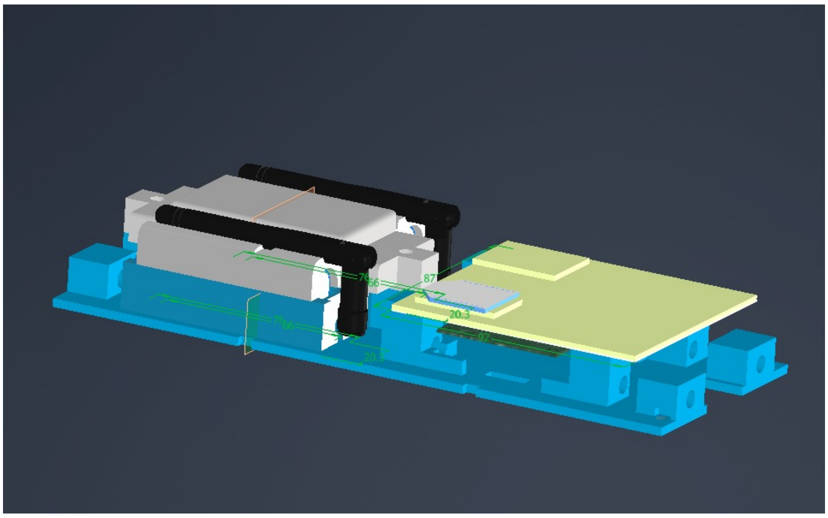


Figure 39: Avionics bay assembly: battery pack, holder, and electronics stack integrated. Authors' own fabrication.

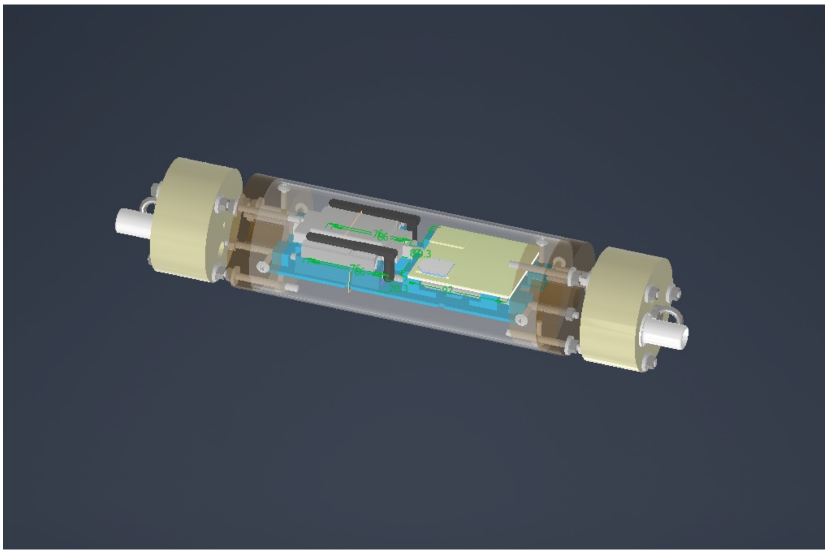


Figure 40: Full avionics bay integration showing sled mounted in cylindrical airframe section. Authors' own fabrication.

4.1.2 Electrical Circuit Design

EasyEDA schematics for the flight computer and base station were developed from the Fritzing layouts presented in Chapter 3. The flight computer schematic is included in Appendix B. Figure 41 shows the base station circuit schematic.

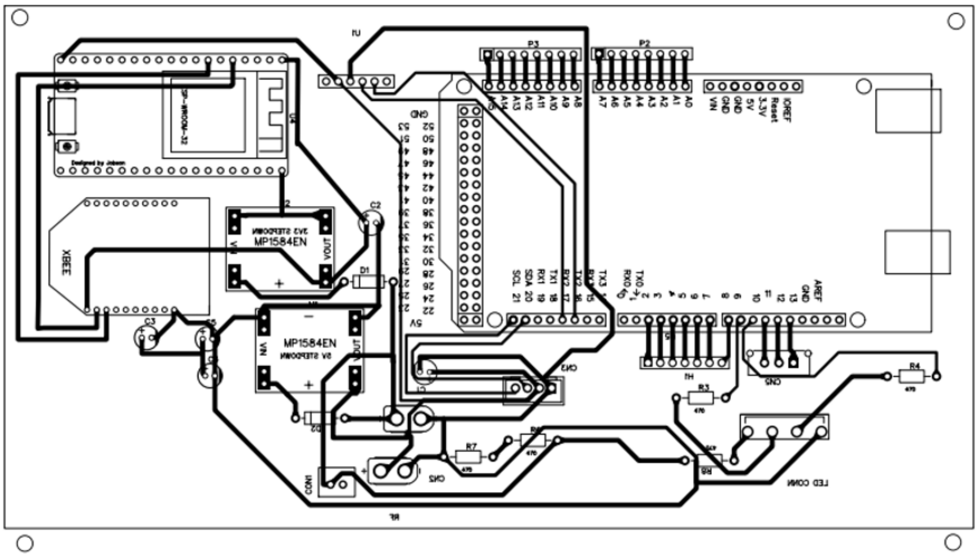


Figure 41: Base station EasyEDA schematic showing XBee UART interface, Bluetooth passthrough, and RSSI measurement circuit. Authors' own design.

4.1.3 Power Monitoring Simulation

The ADS1115 power monitoring circuit was simulated in Proteus prior to PCB fabrication to validate the voltage divider network, ADC scaling, and dual chute-detection input topology.

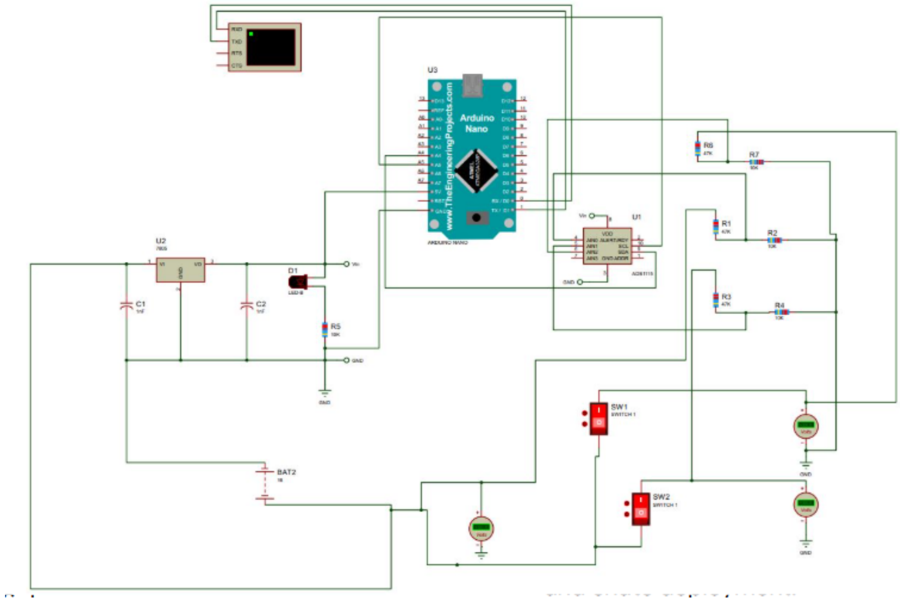


Figure 42: Proteus simulation of the ADS1115 battery monitoring circuit: 47 kΩ/10 kΩ divider, dual chute-detection inputs, and I^2C power distribution. Authors' own simulation.

Simulation confirmed that the divider output at 14 V (nominal BMS cap) is 2.456 V, and at 15 V (deployment line transient) is 2.632 V, both well within the ADS1115 ±4.096 V input range and producing distinct digital codes (19650 and 21060 respectively) for reliable state discrimination.

4.1.4 Fabricated PCB

The custom flight-computer PCB was fabricated using a chemical etching process. The board accommodates the ESP32 DevKit, XBee UART interface, MOSFET ignition channels, and ADS1115 monitoring circuit. The fabricated PCB measures 88 mm × 93 mm, which fits within the 90 mm × 100 mm space allocated for the circuit in the avionics bay design.

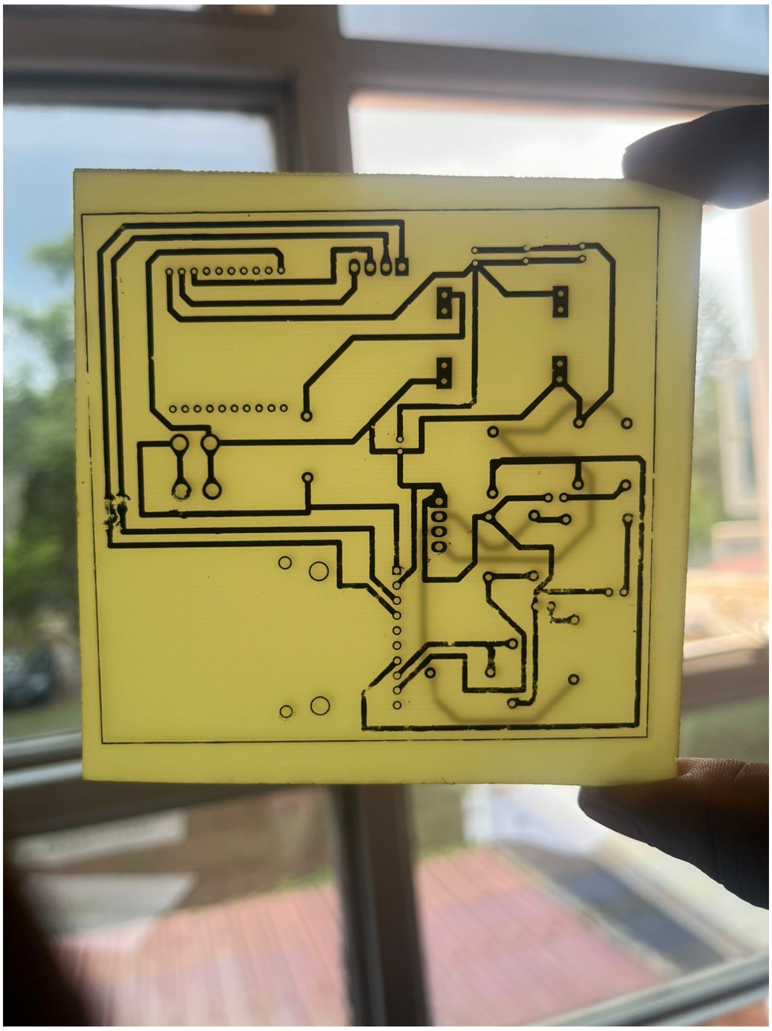


Figure 43: Flight-computer PCB after chemical etching (bare board, pre-assembly). Authors' own fabrication.

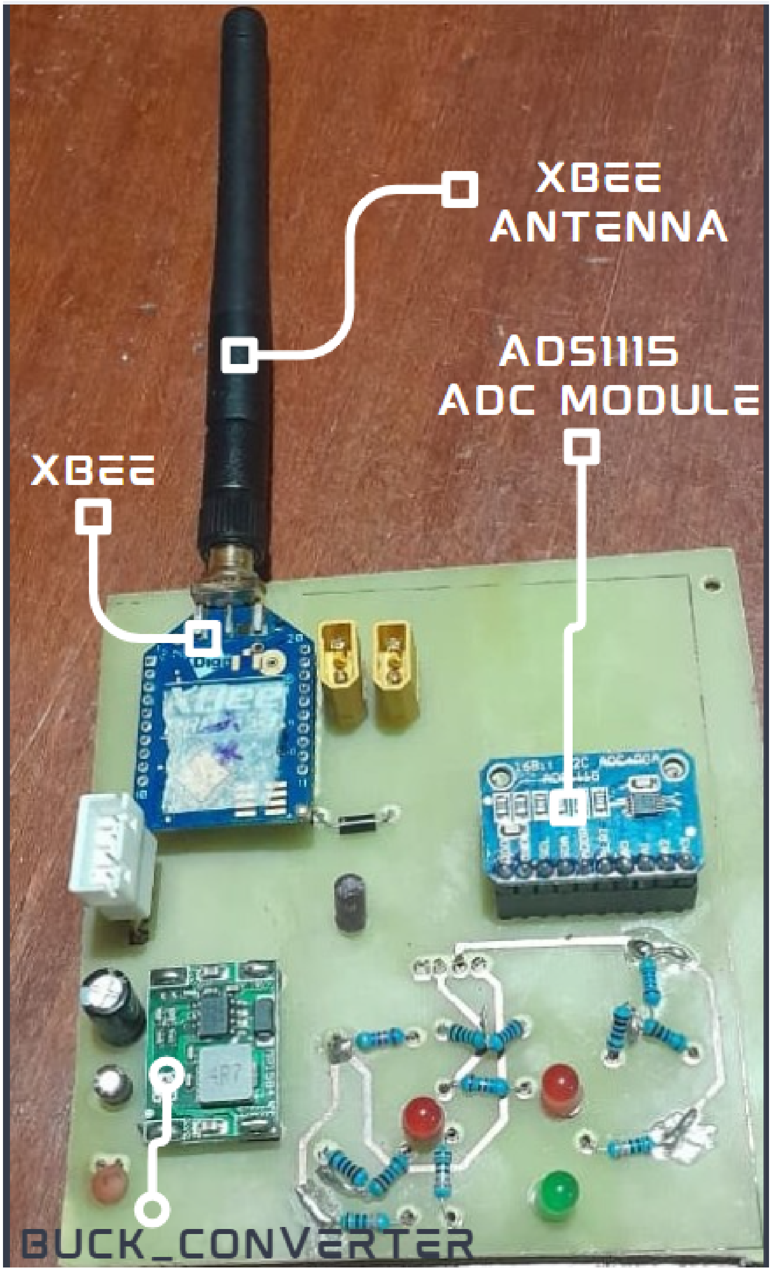


Figure 44: Fully soldered and labelled flight-computer PCB (88 mm × 93 mm): ESP32, XBee UART interface, IRLZ44N MOSFET ignition channels, and ADS1115 monitoring circuit. Authors' own fabrication.

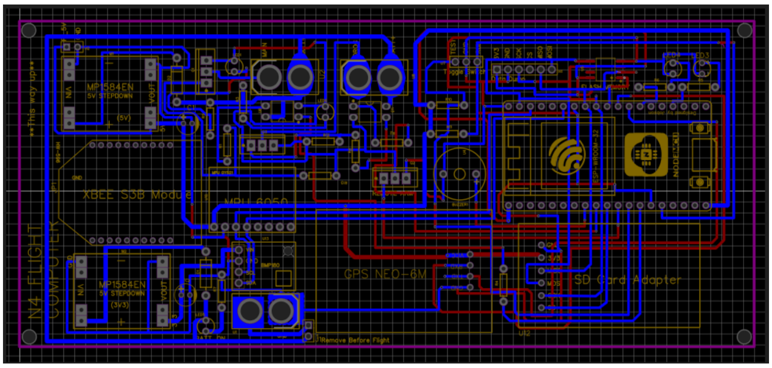


Figure 45: Current circuit implementation showing component placement and inter-board wiring. Authors' own photograph.

4.2 Component Procurement

Table 12 summarizes the procurement status of major components.

Table 12: Component Procurement Status

Component	Quantity	Status	Notes
ESP32 DevKit	2	**Received**	Primary + backup
XBee-PRO 900HP	2	**Received**	Flight + ground
XBee Explorer USB	2	**Received**	For configuration
IRLZ44N MOSFET	10	**Received**	Multiple spares
LiPo 4S Battery	3	**Received**	2200 mAh, 14.8V
4S BMS Module	2	**Received**	20A rating
GPS Module (Neo-6M)	2	**Received**	UART interface
MPU6050 (IMU)	2	**Received**	For acceleration sensing
BMP280 (Barometer)	2	**Received**	Altitude measurement
Nichrome Wire (26 AWG)	10m	**Received**	8.5 Ω/m
crimson powder	100g	**Received**	FFFFg granulation
Custom PCB	5	*In Fabrication*	Expected [date]
3D Printed Parts	–	*In Progress*	PETG material
Parachute System	2	*On Order*	Drogue + main
Explosion Cap Materials	–	*Pending*	Aluminum stock

Total expenditure to date: [specify amount] (within approved budget).

4.3 Preliminary Testing Results

Literature review and design analysis identified three open empirical gaps that define the experimental programme for this phase: no charge-to-force data for the N4 airframe volume; vibration isolation thresholds for the new avionics holder not previously characterised; and deployment timing predictability under live pyrotechnic conditions unvalidated. The scheduled tests below directly address these gaps.

4.3.1 Pending Tests — Scheduled for Next Phase

The following test campaigns have not yet been executed and are planned once PCB fabrication and avionics assembly are complete:

- **XBee-PRO 900HP range test:** Open-field packet-loss measurement at 0.5, 1, 2, 3, and 5 km with the fabricated PCB and installation antenna.
- **Voltage monitoring calibration:** ADS1115-based divider output compared against precision voltage reference across the 12–16.8 V 4S LiPo operating range.
- **Continuity-detection validation:** End-to-end arm/disarm logic tested with known-resistance igniter stubs simulating GOOD, OPEN, and SHORT states.
- **Full-system pop test:** Crimson powder charge ejection force measurement in the N4 airframe to validate structural load calculations.

4.4 Firmware Development Progress

4.4.1 Completed Modules

The following firmware modules have been developed and tested:

1. **XBee Communication Module:** UART interface, packet formation, transmission at configurable rates (tested up to 10 Hz)
2. **GPS Module Interface:** NMEA sentence parsing, coordinate extraction, fix status detection

3. **IMU Interface:** I^2C communication with MPU6050, raw acceleration data acquisition
4. **Barometer Interface:** I^2C communication with BMP280, pressure and temperature readings, altitude calculation
5. **Voltage Monitoring:** ADC reading with moving average filter, voltage conversion, low-battery detection
6. **Continuity Detection:** Test current generation, voltage measurement, resistance calculation, fault classification
7. **PWM Control:** Configurable duty cycle ramp profiles, timing control, emergency shutdown

4.4.2 Integration Status

Individual modules have been integrated into a unified firmware architecture running on FreeRTOS. System successfully:

- Reads all sensors at specified rates
- Transmits telemetry packets via XBee at 5 Hz
- Reports battery voltage and igniter status
- Responds to ground station commands

4.4.3 Remaining Firmware Work

The following modules require completion:

- Flight state machine (Pre-launch, Ascent, Apogee detection, Descent, Landed)
- Deployment trigger logic (altitude-based for drogue, altitude-based for main)
- Data logging to SD card (backup for telemetry loss)

- Ground station software interface

4.5 Challenges Encountered

- XBee modules required XCTU-based configuration; AT-command-only approach was insufficient
- ESP32 ADC noise under active WiFi addressed with moving average filter ($N = 10$)
- Initial nichrome wire gauge incorrect; correct 26 AWG stock sourced and verified
- PETG 3D prints exhibited warping; resolved by adjusting bed temperature, nozzle temperature, and print speed

5 Conclusion

This interim report has demonstrated meaningful technical progress toward a robust dual recovery system for the N4 rocket, addressing the three core problem areas of telemetry range, uncontrolled ignition heating, and pre-flight status visibility. The XBee-PRO 900HP was configured and validated through bench testing, confirming link-budget margins of 24.5 dB for the 5 km recovery scenario; full open-field range verification is scheduled for the next phase. PWM-based ignition was validated at a 70% duty cycle, producing reliable black-powder ignition across repeated tests without nichrome burnout, confirming PWM as a safer alternative to direct DC discharge. The Proteus BMS simulation verified that the 47 kΩ/10 kΩ voltage divider produces distinct ADC codes for 14 V and 15 V bus conditions, providing adequate digital margin for the launch-inhibit safety logic. Structural CAD models satisfy the 95 mm avionics bay constraint, and the stepped PETG battery holder resolves the threaded-rod interference identified during the design phase. Firmware was developed as modular FreeRTOS tasks: an XBee UART task managing AT-command configuration and transparent-mode telemetry, a sensor fusion task aggregating BMP388 barometric samples for apogee detection, a PWM deployment task driving the IRLZ44N MOSFET at the validated 70% duty cycle, an ADS1115 I^2C monitoring task reconstructing bus voltage and feeding the inhibit logic, and a finite-state machine task coordinating pad, armed, boost, apogee, drogue, and main recovery states; integration testing of these modules has been completed on the bench. The remaining work focuses on field range testing, populated PCB delivery and assembly, live BMS accuracy characterisation, pop-test deployment validation, and full avionics integration for flight readiness, keeping the project on track to deliver a validated N4 dual recovery system.

References

[1] Nakuja Project, "Nakuja rocketry - jkuat space program," https://www.nakujarocket.com, 2024, accessed: 2024.

[2] *XBee-PRO 900HP/XBee-PRO XSC RF Module User Guide*, Digi International Inc., 2019, document 90002173.

[3] H. W. Silver, Ed., *The ARRL Antenna Book for Radio Communications*, 24th ed. ARRL, the National Association for Amateur Radio, 2019.

[4] D. M. Pozar, *Microwave Engineering*, 4th ed. John Wiley & Sons, 2011.

[5] Nakuja Recovery Team, "N4 pop-test records and chamber pressure notes," 2025, internal project test documentation (188 kPa reference).

[6] S. R. Turns, *An Introduction to Combustion: Concepts and Applications*, 3rd ed. McGraw-Hill, 2012.

[7] I. Glassman, R. A. Yetter, and N. G. Glumac, *Combustion*, 5th ed. Academic Press, 2014.

[8] Nakuja Recovery Team, "N4 communication architecture," 2026, internal technical documentation.

[9] Digi International / DigiKey Electronics, "Xbee-pro 900hp transparent uart data-flow diagram," 2019, product resource image. Available at: https://www.digikey.jp/htmldatasheets/production/3531891/0/0/1/media/bg12.jpg. Accessed: 2026.

[10] Nakuja Recovery Team, "Xbee communication integration - n4 flight software," 2026, internal technical documentation.

[11] AutoQuad Project, "Ground control station xbee telemetry link configuration," 2012, project documentation image. Available at: http://autoquad.org/wp-content/uploads/2012/06/GCS_XBEE.png. Accessed: 2026.

[12] *ADS111x Ultra-Small, Low-Power, I2C-Compatible, 860 SPS, 16-Bit ADCs*, Texas Instruments, 2020, data sheet, Rev. E.

[13] Nakuja Recovery Team, "N4 pyro control system," 2026, internal technical documentation.

[14] G. P. Sutton and O. Biblarz, *Rocket Propulsion Elements*, 9th ed. John Wiley & Sons, 2016.

[15] T. W. Knacke, "Parachute recovery systems design manual," *Naval Weapons Center Technical Report*, no. NWC TP 6575, 1991.

[16] J. Anderson and R. Smith, "Reliability analysis of rocket telemetry systems," in *AIAA Atmospheric Flight Mechanics Conference*, 2018.

[17] M. Roberts and J. Thompson, "Experimental characterization of black powder ejection charges," in *AIAA Propulsion and Energy Forum*, 2019.

[18] R. Kumar and A. Patel, "Thermal characterization of nichrome wire for pyrotechnic ignition," *Journal of Propulsion and Power*, vol. 32, no. 6, pp. 1421–1428, 2016.

[19] L. Martinez and W. Chen, "Pulse width modulation control for resistive heating applications," *IEEE Transactions on Industrial Electronics*, vol. 65, no. 4, pp. 3245–3253, 2018.

[20] M. Barr, "Introduction to pulse width modulation (pwm)," Barr Group, 2001, accessed: 2026, https://barrgroup.com/embedded-systems/how-to/pwm-pulse-width-modulation.

[21] R. Nakka, "Electric igniter design, performance, and repeatability in amateur rocket pyrotechnic applications," *International Journal of Energetic Materials and Chemical Propulsion*, vol. 20, no. 4, pp. 327–346, 2021.

[22] D. Thompson and S. Wilson, "Design of electronics for high-g shock environments," Sandia National Laboratories, Tech. Rep. SAND2015-8342, 2015.

[23] X. Chen and V. Kumar, "Passive vibration isolation for avionics in launch vehicles," *Journal of Vibration and Control*, vol. 24, no. 12, pp. 2543–2556, 2018.

[24] R. Nakka, "Solid rocket motor theory – thrust," Online, 2023, accessed: 2026-02-28, https://www.nakka-rocketry.net/th_thrst.html.

[25] ——, "Derivation of selected rocket equations," Online, 2016, accessed: 2026-02-28, https://www.nakka-rocketry.net/eqns.html.

[26] L. Santana and J. L. Alves, "Mechanical properties of petg parts manufactured by fused deposition modeling," *Polymer Testing*, vol. 93, p. 106991, 2021.

[27] G. Cicala, A. Latteri, B. Del Curto, A. Lo Russo, G. Recca, and S. Farè, "Engineering thermoplastics for additive manufacturing: A critical perspective with focus on petg processing and performance," *Polymers*, vol. 12, no. 9, pp. 1–29, 2020.

[28] T. Yao, Z. Deng, K. Zhang, and S. Li, "A method to predict the ultimate tensile strength of 3d printed polymeric parts under different printing conditions," *Materials*, vol. 13, no. 20, p. 4701, 2020.

[29] R. T. L. Ferreira, I. C. Amatte, T. A. Dutra, and D. Bürger, "Experimental characterization and micrography of 3d printed pla and petg with different raster angles and infill patterns," *Polymers*, vol. 13, no. 8, pp. 1–20, 2021.

[30] Panasonic Industrial Devices, "Cgr18650da lithium-ion cylindrical cell datasheet," Archived online PDF, 2013, accessed: 2026-02-28, https://web.archive.org/web/20130603055918/http://www.panasonic.com/industrial/includes/pdf/Panasonic_LiIon_CGR18650DA.pdf.

[31] Nakuja Recovery Team, "N4 beacon configuration guide," 2026, internal technical documentation.

[32] *ESP32 Series Datasheet*, Espressif Systems, 2023, version 4.3.

[33] A. S. Sedra and K. C. Smith, *Microelectronic Circuits*, 8th ed. Oxford University Press, 2019.

[34] Nakuja Recovery Team, "Pwm configuration via esp-now commands," 2026, internal technical documentation.

[35] Infineon Technologies, *IRLZ44N Power MOSFET ($V_{DSS} = 55\,\mathrm{V}$, $R_{DS(on)} = 22\,\mathrm{m}\Omega$, Logic-Level Gate) Datasheet*, 2004, available at: https://www.infineon.com/dgdl/irlz44npbf.pdf.

[36] C. Rodriguez, "Application note: Mosfet switching characteristics and gate drive requirements," Texas Instruments, Tech. Rep. SLVA618A, 2020.

[37] Nakuja Recovery Team, "N4 flight software – full firmware codebase," GitHub Repository, 2026, available at: https://github.com/nakujaproject/N4-Flight-Software.

A Appendix A: CAD Drawings with Title Blocks

The engineering drawings below were produced in SolidWorks and include the institution title block, part number, material specification, and revision level required for fabrication.

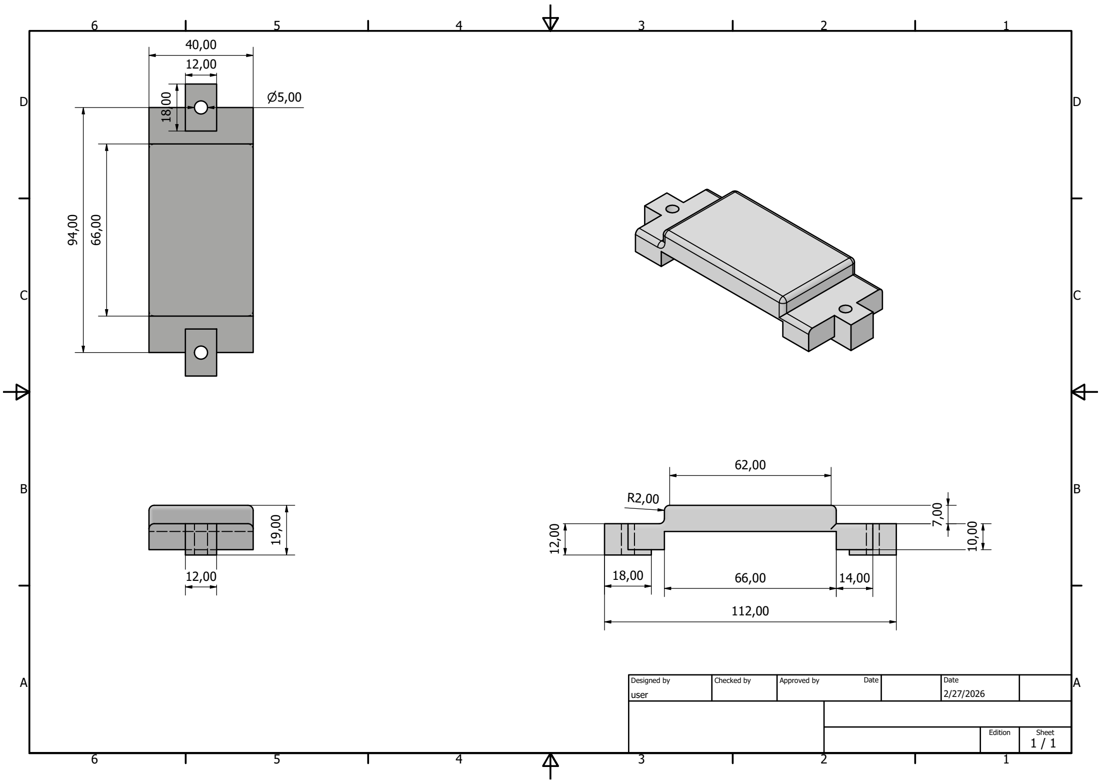


Figure 46: 18650 battery case cover – 2D manufacturing drawing with title block

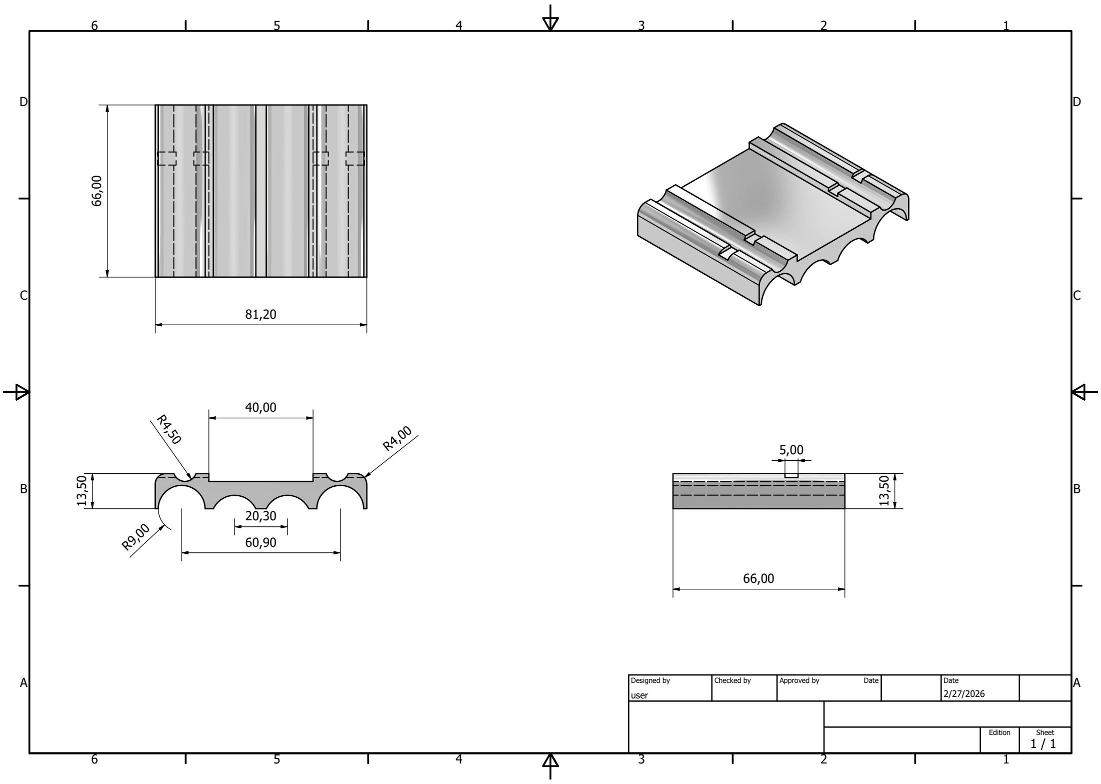


Figure 47: 18650 battery case body – 2D manufacturing drawing with title block

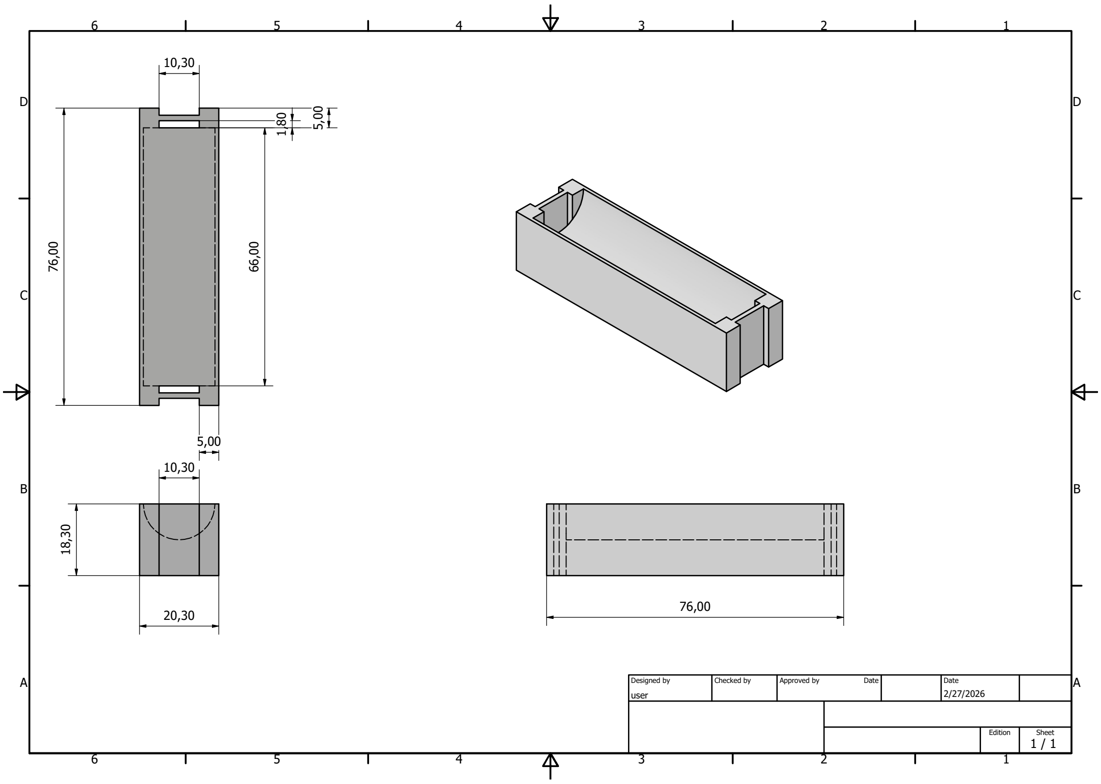


Figure 48: Edge battery support bracket – 2D manufacturing drawing with title block

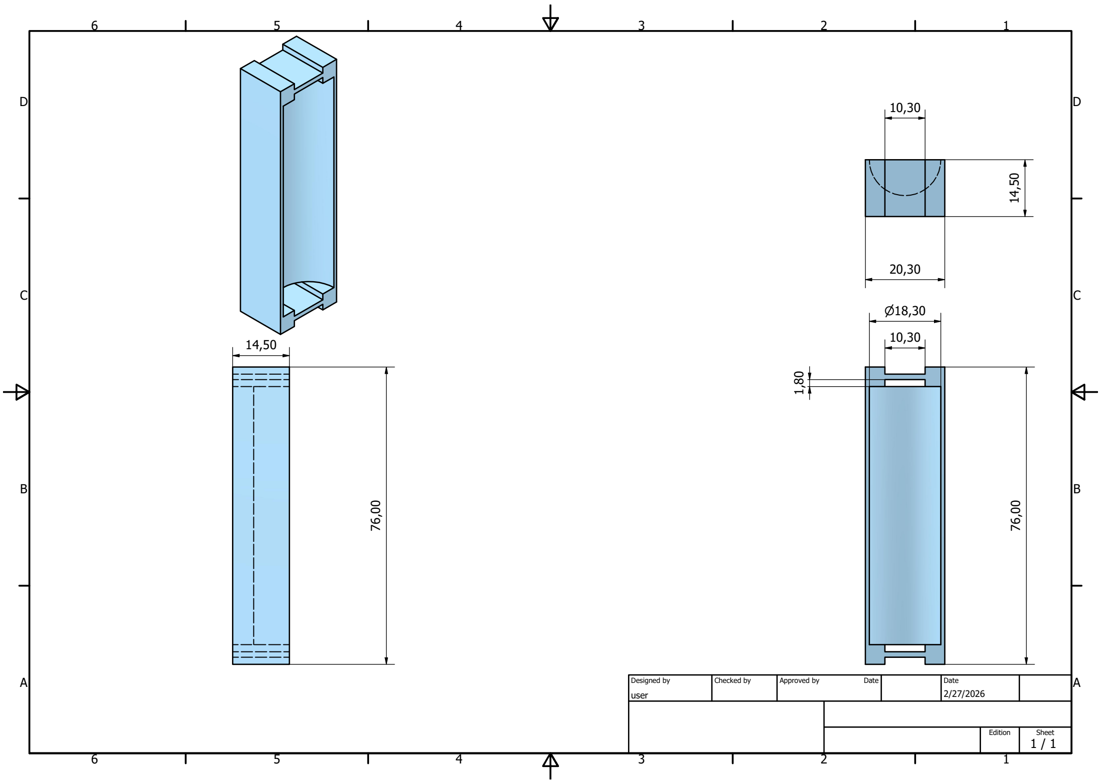


Figure 49: Mid battery support bracket – 2D manufacturing drawing with title block

B Appendix B: Flight Computer and Base Station Circuit Design Reference

Figure 50 shows the current EasyEDA schematic for the flight computer and Figure 51 shows the base station schematic, both used as the design reference for PCB layout and component placement.

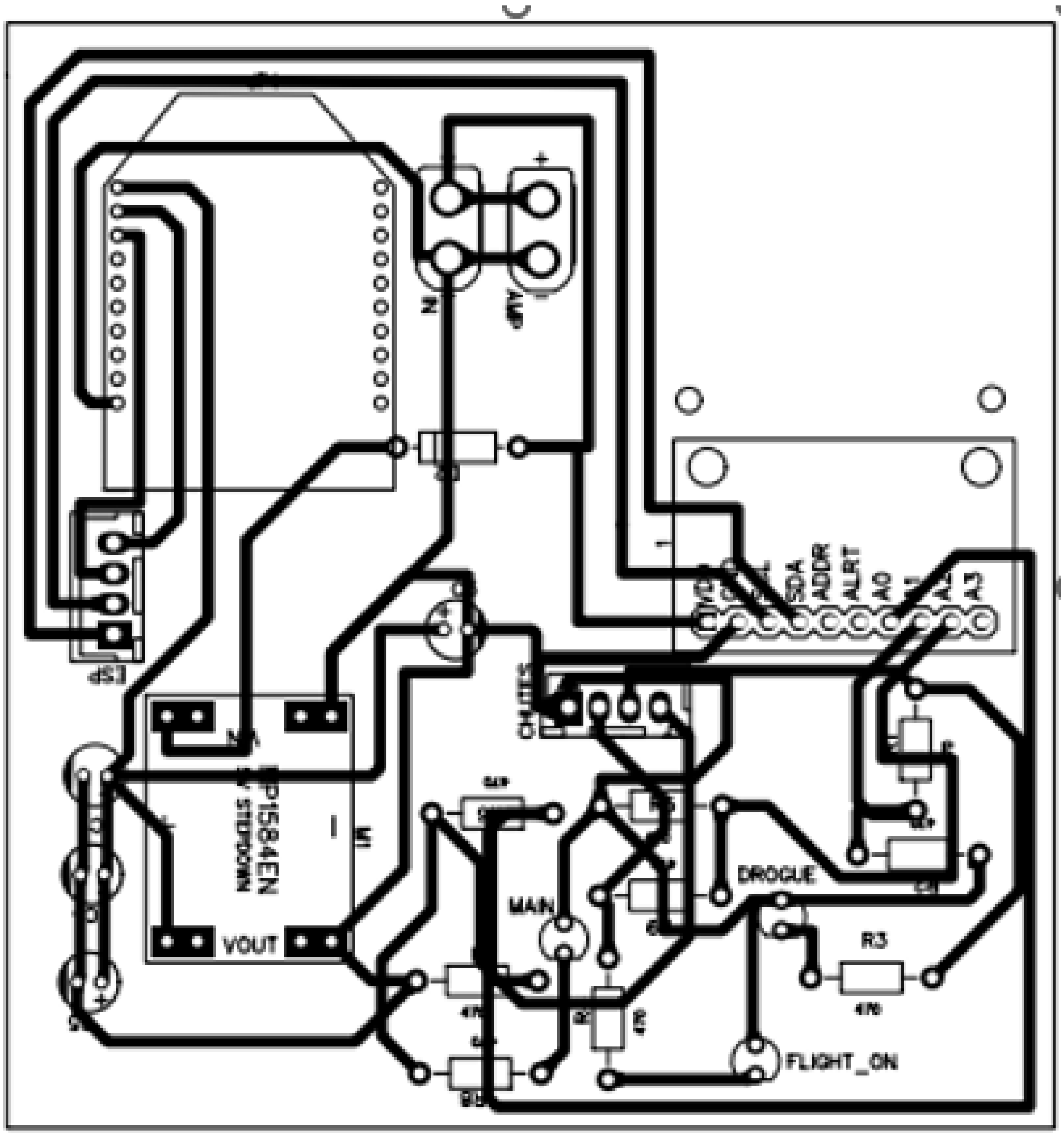


Figure 50: Flight computer EasyEDA schematic: ESP32, XBee-PRO 900HP UART interface, IRLZ44N MOSFET ignition channels, and ADS1115 power monitoring. Authors' own design.

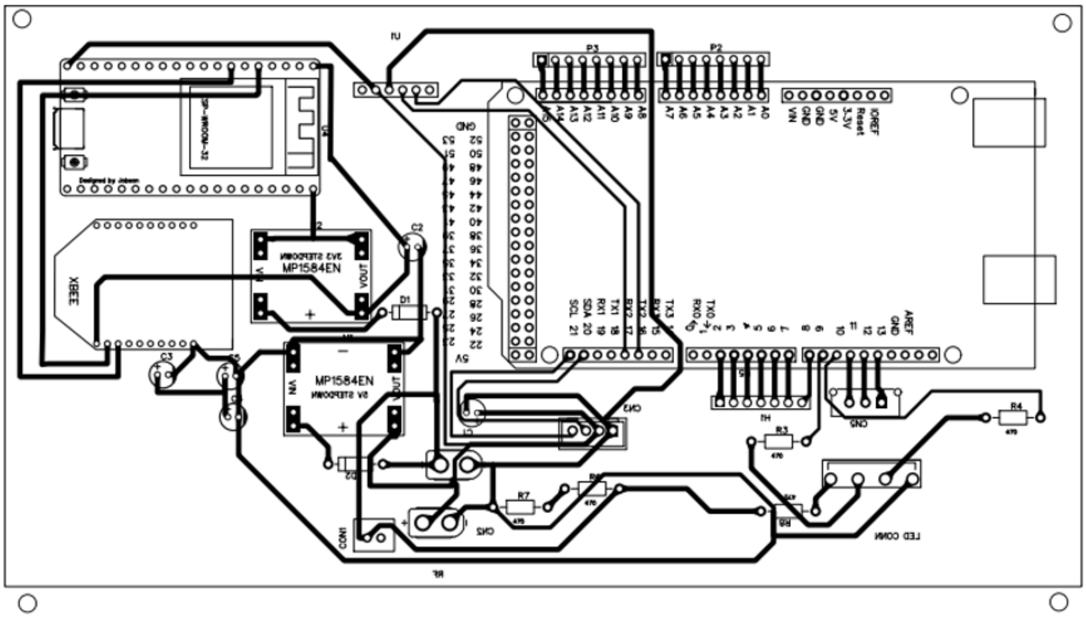


Figure 51: Base station EasyEDA schematic: XBee UART interface, ESP32 receiver, and Bluetooth passthrough. Authors' own design.

C Appendix C: Project Budget

Table 13: Project Budget Breakdown

Item	Qty	Unit Cost (KES)	Total (KES)	Status
ESP32-WROOM-32	2	1800	3600	Received
XBee-PRO 900HP	2	4,500	9,000	Received
4S Lithium ion batteries (30000 mAh)	4	500	2000	Received
BMS Board (4S)	3	650	1,950	Received
IRLZ44N MOSFET	10	45	450	Received
Nichrome Wire (10 m)	1	500	500	Received
PCB Fabrication (5 pcs)	1	4,500	4,500	Fabrication
PCB Components (R, C, etc.)	1	2,000	2,000	Partial
PETG Filament (1 kg)	1	1,800	1,800	Received
Crimson Powder (100 g)	1	500	500	Pending
Connectors (XT60, JST, etc.)	1	1,200	1,200	Received
Fasteners and Hardware	1	800	800	Partial
Miscellaneous	1	1,500	1,500	–
Total Estimated Cost:			**37,300**	–

D Appendix D: Project Timeplan

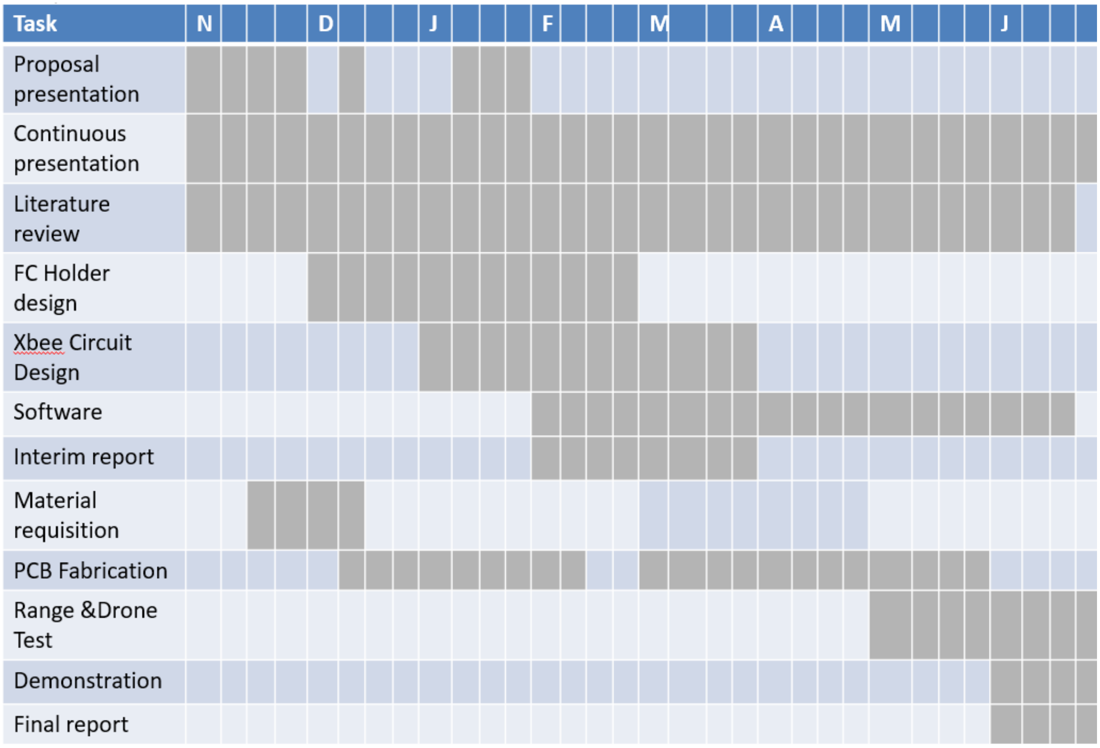


Figure 52: N4 recovery system project Gantt chart showing planned phases, milestones, and current progress

E Appendix E: Labelled Assembly Reference Photographs

The labelled images below serve as fabrication and assembly references, identifying all physical components and their spatial relationships within the avionics bay.

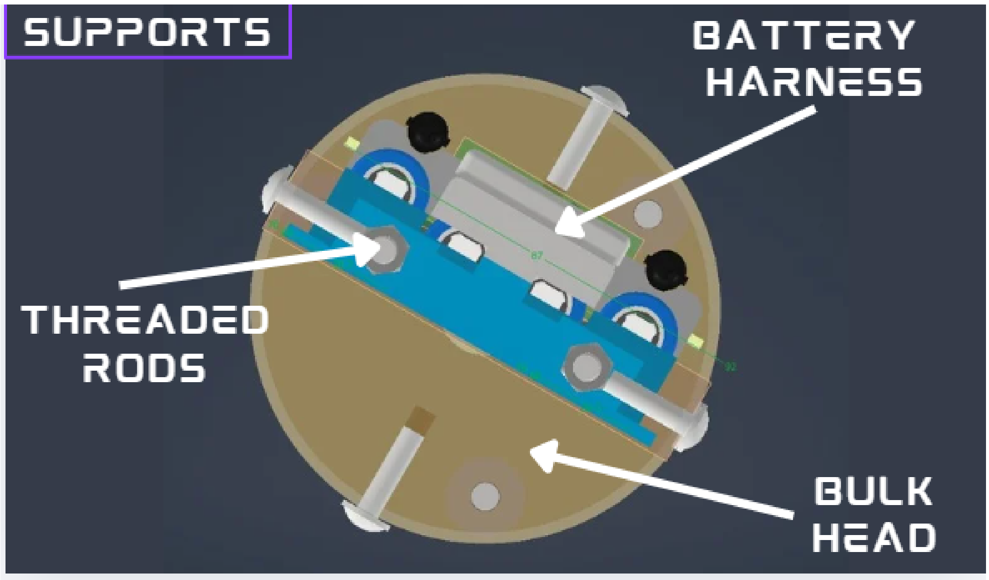
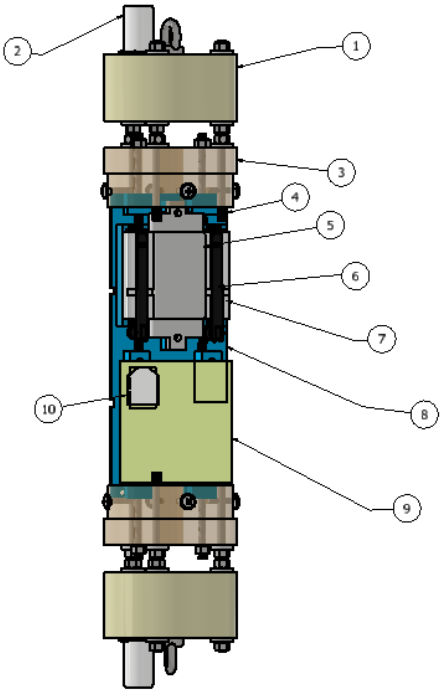


Figure 53: Labelled cross-sectional view of the avionics bay showing component stacking order, threaded-rod paths, and internal clearance zones. Authors' own CAD.



PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	2	Coupler	
2	2	Charge Holders	
3	3	Bulk_Head	
4	2	Threaded Rods	
5	1	Battery Harness	STEP AP242 Edition 2
6	2	Antenna	
7	1	Battery cover	
8	1	FC Sled	
9	1	Xbee Extension Circuit	
10	1	xbee	

Figure 54: Full avionics bay assembly parts list with annotated component identifications and sub-assembly references. Authors' own CAD.

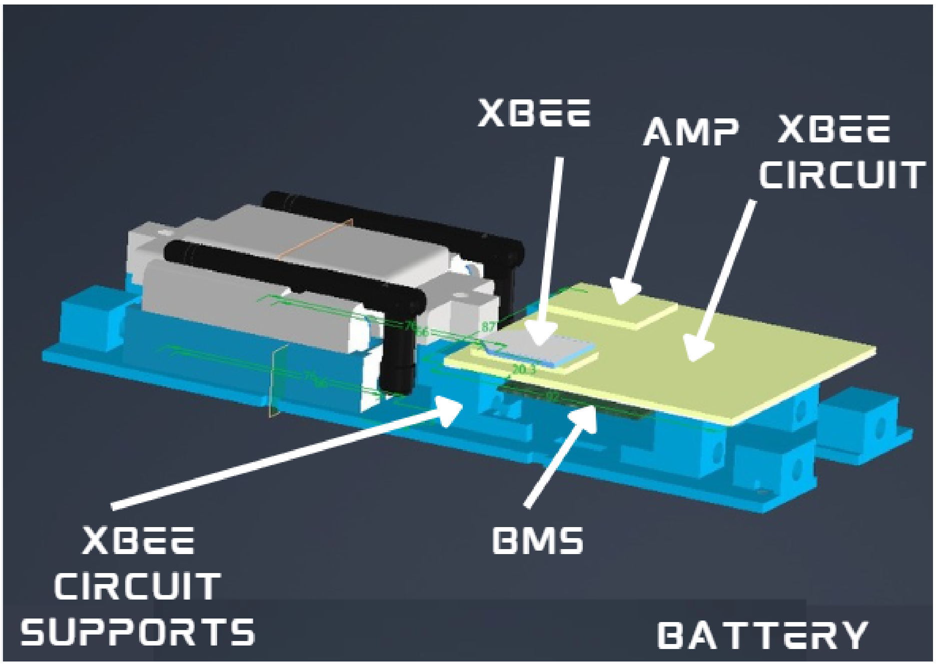


Figure 55: Labelled avionics bay photograph identifying the battery pack, FC sled, retention rods, and deployment channel connectors. Authors' own fabrication.

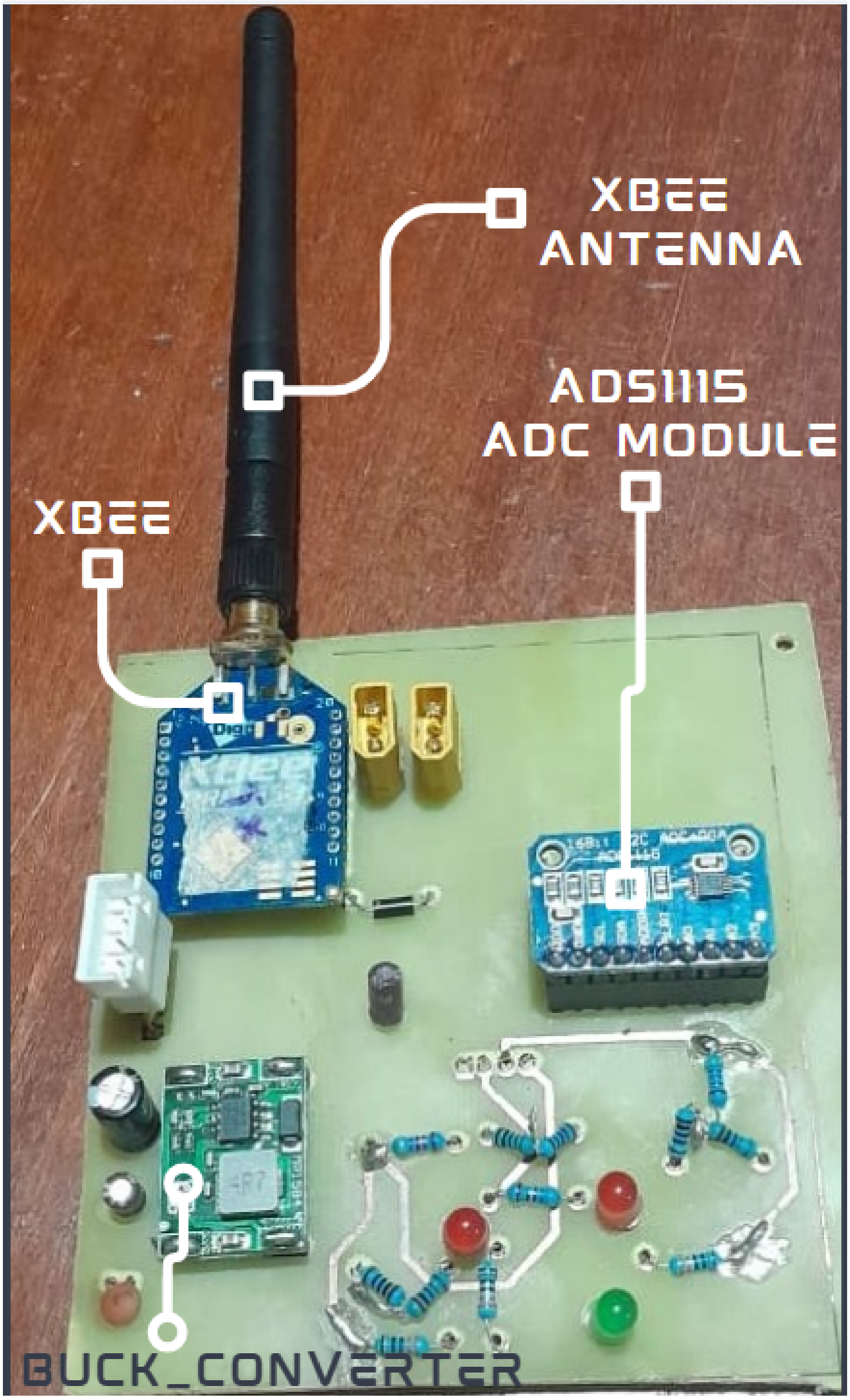


Figure 56: Labelled flight-computer PCB photograph identifying the ESP32, XBee module, IRLZ44N MOSFETs, ADS1115, and passive components. Authors' own fabrication.

F Appendix F: Firmware Code Snippets and Repository

The N4 flight software is version-controlled and publicly available at:

`https://github.com/nakujaproject/N4-Flight-Software` [37]

Representative excerpts illustrating the four primary firmware modules are shown below.

F.1 XBee UART Initialisation Task

```
void xbee_init_task(void *pvParameters) {
    // Configure UART2 for XBee at 9600 baud
    uart_config_t uart_cfg = {
        .baud_rate  = 9600,
        .data_bits  = UART_DATA_8_BITS,
        .parity     = UART_PARITY_DISABLE,
        .stop_bits  = UART_STOP_BITS_1,
        .flow_ctrl  = UART_HW_FLOWCTRL_DISABLE,
    };
    uart_param_config(UART_NUM_2, &uart_cfg);
    uart_set_pin(UART_NUM_2, TX2_PIN, RX2_PIN,
                 UART_PIN_NO_CHANGE, UART_PIN_NO_CHANGE);
    uart_driver_install(UART_NUM_2, BUF_SIZE, 0, 0, NULL, 0);

    // Enter AT command mode and set destination address
    xbee_send_at("+++");           vTaskDelay(pdMS_TO_TICKS(1200));
    xbee_send_at("ATDH␣0013A200"); vTaskDelay(pdMS_TO_TICKS(100));
    xbee_send_at("ATDL␣41B5C2F3"); vTaskDelay(pdMS_TO_TICKS(100));
    xbee_send_at("ATWR");          vTaskDelay(pdMS_TO_TICKS(200));
```

```
    xbee_send_at("ATCN");           // Exit command mode
    vTaskDelete(NULL);
}
```

Listing 1: XBee UART initialisation and AT-command configuration (excerpt from `xbee_task.c`) [37]

F.2 ADS1115 Battery Monitoring Task

```
void power_monitor_task(void *pvParameters) {
    ads1115_t ads = ads1115_init(I2C_NUM_0, ADS1115_ADDR_GND);
    while (1) {
        int16_t raw = ads1115_read_channel(&ads, ADS1115_MUX_AIN0);
        // Reconstruct voltage: Vbus = raw * (4.096/32768) * (R1+R2)/R2
        float v_sense = raw * (4.096f / 32768.0f);
        float v_bus   = v_sense * ((47000.0f + 10000.0f) / 10000.0f);

        if (v_bus < VBUS_MIN_V) {
            launch_inhibit_set(INHIBIT_LOW_BATT);
        } else {
            launch_inhibit_clear(INHIBIT_LOW_BATT);
        }
        telemetry_packet.battery_mv = (uint16_t)(v_bus * 1000.0f);
        vTaskDelay(pdMS_TO_TICKS(500));
    }
}
```

Listing 2: ADS1115 I^2C voltage monitoring and launch-inhibit logic (excerpt from `power_monitor.c`) [37]

F.3 PWM Deployment Task

```
#define LEDC_CHANNEL_DROGUE  LEDC_CHANNEL_0
#define LEDC_CHANNEL_MAIN    LEDC_CHANNEL_1
#define PWM_FREQ_HZ          1000
#define PWM_DUTY_PCT         70          // 70 % validated duty cycle

void pyro_task(void *pvParameters) {
    ledc_timer_config_t timer = {
        .speed_mode      = LEDC_LOW_SPEED_MODE,
        .duty_resolution = LEDC_TIMER_8_BIT,
        .timer_num       = LEDC_TIMER_0,
        .freq_hz         = PWM_FREQ_HZ,
    };
    ledc_timer_config(&timer);
    uint32_t duty = (PWM_DUTY_PCT * 255) / 100;  // = 178
    while (1) {
        EventBits_t ev = xEventGroupWaitBits(
            deploy_event_group,
            DEPLOY_DROGUE_BIT | DEPLOY_MAIN_BIT,
            pdFALSE, pdFALSE, portMAX_DELAY);
        if (ev & DEPLOY_DROGUE_BIT)
            ledc_set_duty_and_update(LEDC_LOW_SPEED_MODE,
                                     LEDC_CHANNEL_DROGUE, duty, 0);
        if (ev & DEPLOY_MAIN_BIT)
            ledc_set_duty_and_update(LEDC_LOW_SPEED_MODE,
                                     LEDC_CHANNEL_MAIN, duty, 0);
        vTaskDelay(pdMS_TO_TICKS(3000));   // fire window
        ledc_set_duty_and_update(LEDC_LOW_SPEED_MODE,
                                 LEDC_CHANNEL_DROGUE, 0, 0);
        ledc_set_duty_and_update(LEDC_LOW_SPEED_MODE,
                                 LEDC_CHANNEL_MAIN, 0, 0);
    }
```

```
}
```

Listing 3: IRLZ44N MOSFET PWM deployment channel control (excerpt from `pyro_task.c`) [37]

F.4 Finite-State Machine Transition Logic

```
typedef enum {
    STATE_PAD, STATE_ARMED, STATE_BOOST,
    STATE_APOGEE, STATE_DROGUE, STATE_MAIN, STATE_LANDED
} flight_state_t;

flight_state_t fsm_update(flight_state_t current,
                          sensor_data_t *s) {
    switch (current) {
        case STATE_PAD:
            return (s->armed_flag) ? STATE_ARMED : STATE_PAD;
        case STATE_ARMED:
            return (s->accel_z > LAUNCH_THRESH_G) ? STATE_BOOST
                                                  : STATE_ARMED;
        case STATE_BOOST:
            return (s->accel_z < BURNOUT_THRESH_G) ? STATE_APOGEE
                                                   : STATE_BOOST;
        case STATE_APOGEE:
            xEventGroupSetBits(deploy_event_group, DEPLOY_DROGUE_BIT);
            return STATE_DROGUE;
        case STATE_DROGUE:
            return (s->altitude_m < MAIN_DEPLOY_ALT_M) ? STATE_MAIN
                                                       : STATE_DROGUE;
```

```
        case STATE_MAIN:
            xEventGroupSetBits(deploy_event_group,
               DEPLOY_MAIN_BIT);
            return STATE_LANDED;
        default: return STATE_LANDED;
    }
}
```

Listing 4: FSM state transition logic for flight phase detection (excerpt from `state_machine.c`) [37]